TSH GROUPED BIOGAS RECOVERY PROJECT

TSH RESOURCES BERHAD

Document Prepared by TSH Resources Berhad

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Grouped Project Description

TSH Grouped Biogas Recovery Project (hereinafter referred to as the project) is a grouped project which consists of biogas recovery projects located in Indonesia. The project is owned by TSH Resources Berhad (hereinafter referred to as the project owner), and implementation of the projects are by the individual plant operators of each PAI (project activity instance).

The purpose of this Grouped Project is to recover the biogas generated from the wastewater (with or without sludge) treatment process which would have otherwise, be emitted into the atmosphere resulting in GHG emissions.

The recovered biogas is expected to be utilized for (a) electricity generation for own consumption, and/or (b) upgrading of biogas and bottling for distribution to end user, and/or distributing to natural gas distribution grid, and with the option of flaring for any excess quantity.

Project Activity Instance Description

For the start, the project will consist of four (4) project activity instances as follows:

Table 1: Project Activity Instance (PAIs) Description

No	Project activity instance	Geographical Coordinates	Average FFB Processed (t/y)[1]	Average POME Generated (m^3/y)[1]
PAI1	PT Sarana Prima Multi Niaga (SPMN)	S 1°57'11.50" E 112°55'41.08"	234,180	154,487
PAI 2	PT Andalas Agro Industry	S 00 03' 21"	149,430	104,982

[1] Three (3) years historical record (2020 – 2022) obtained from the specific mills

No	Project activity instance	Geographical Coordinates	Average FFB Processed (t/y)[1]	Average POME Generated (m^3/y)[1]
	(AAI)	E 99 58' 21"		
PAI 3	PT Farinda Bersaudara (FDB)	S 0°41'43.23" E 116°15'41.98"	298,803	206,174
PAI 4	PT Andalas Wahana Berjaya (AWB)	S 1° 7'12.26" E 101°33'5.48"	134,617	144,647

Baseline Scenario

In the processing of FFB, wastewater commonly known as Palm Oil Mill Effluent (POME) is generated. POME is well known as a high strength wastewater with very high organic and oil content. For the treatment of POME during the baseline scenario, the POME is distributed into open anaerobic lagoons for anaerobic treatment. After the anaerobic treatment, the treated POME will be discharged for either land irrigation or river.

Project Scenario

The project involves the improvement of POME treatment by introducing anaerobic digester, a membrane is installed over the lagoon to capture the methane emitted. After the digesters, the residue will be further treated through a series of aerobic processes before finally irrigated to land or discharged to river, as per baseline. Biogas generated from the digesters will be captured and utilized. Biogas will be converted and used for various purpose:

a. To use internally
b. To upload to grid
c. Excess biogas will be flared.

The estimated annual average and total GHG emission reductions during the first seven (7) year crediting period of the PAI is tabulated below:

No	Project activity instance	Annual *Average GHG Emission (tCO2e)*	*Total* GHG emission reductions for first crediting period *(tCO_2e)*
PAI1	PT Sarana Prima Multi Niaga (SPMN)	30,031	240,251

No	Project activity instance	Annual *Average GHG Emission (tCO2e)*	*Total* GHG emission reductions for first crediting period *(tCO₂e)*
PAI 2	PT Andalas Agro Industry (AAI)	21,115	168,919
PAI 3	PT Farinda Bersaudara (FDB)	26,150	209,198
PAI 4	PT Andalas Wahana Berjaya (AWB)	25,877	207,020
	Total (tCO_{2e})	**103,174**	**825,388**

1.2 Sectoral Scope and Project Type

The project activity under consideration is a grouped project activity. The project activity instances as part of the grouped project are defined as follow:

Table 2: Sectoral Scope and Project Type

Sectoral Scope	13 - Waste handling and disposal
Type	III - Other project activities
Category	AMS-III.H (Version 19.0) - Methane recovery in wastewater treatment

Sectoral Scope	1 - Energy industries (renewable/non-renewable sources)
Type	I - Renewable energy projects
Category	AMS-I.D (Version 18.0) - Grid connected renewable electricity generation AMS-1.F (Version 05.0) – Renewable electricity generation for captive use and mini-grid

Based on sectoral scope mentioned above, the following combinations of technologies/measures will be covered under this Grouped Project:

Table 3: Combinations of technologies/measures covered in Grouped Project

No	Measure/ Technology	Methodology Combination
1	Installation of anaerobic covered lagoon/digesters for treatment of wastewater and recovery of biogas. The biogas is utilized for captive use electricity, upload to grid, and any excess biogas is flared.	AMS-III.H, version 19.0 AMS-I.D, version 18.0 AMS-I.F, version 5.0
2	Installation of anaerobic covered lagoon/digesters for treatment of wastewater and recovery of biogas. The biogas is utilized for captive use electricity. Any excess biogas is flared.	AMS-III.H, version 19.0 AMS-I.F, version 5.0

1.3 Project Eligibility

The project activity involves the implementation of anaerobic wastewater treatment system and the combustion of biogas. Therefore, as per Section 2.1 of the VCS Standard, version 4.4, the project activity is eligible under the scope of the VCS Program

1.4 Project Design

☒ The project is a grouped project.

Eligibility Criteria

The project is a grouped project and following are the eligibility criteria for inclusion of new project activity instances into the grouped project activity:

- **Applicability Conditions:** The project activity instances shall meet applicability conditions for applicable methodology (AMS III.H version 19.0, AMS I.D version 18.0 and AMS I.F version 05.0 as defined in section 3.2
- **Geographical Area:** The project activity instance locates in the geographic boundary for the grouped project, i.e., Indonesia, and owned by the project owner or the subsidiary of the project owner.
- **Baseline scenario:** All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 3.4

- **Technology type:** The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mills using either anaerobic digester or covered anaerobic lagoon, and gas engine for power generation.
- **Additionality:** The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology. As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types that are defined as automatically additional if the PAIs apply such technologies and meet specified conditions. The positive list of technology including the methane recovery in wastewater treatment where the specific conditions to be met is listed below:
    - The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation;
    - There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites;
    - There is no capacity increase in the wastewater treatment system;
    - No other alternative economic activity is expected to be undertaken on the land of the existing lagoon;
    - The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW.
- **Start Date:** The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.

The following table demonstrates how the initial project activity instances being included in the grouped project activity fulfil above mentioned eligibility conditions:

Table 4: PAIs Eligibility Criteria

No	Eligibility Criteria	PAIs Eligibility
1	**Applicability Conditions:** The project activity instances shall meet applicability conditions for applicable methodology (AMS III.H version 19.0 and AMS I.D version 18.0 and/ or AMS-I.F version 05.0	All initial project activity instances meet respective applicability conditions of methodology AMS III.H, version 19.0 and AMS I.F Version 05.0 and/or AMS I.D Version 18.0. Hence this eligibility criteria are fulfilled. Please refer section 3.2 of this

No	Eligibility Criteria	PAIs Eligibility
		document for applicability
2	**Geographical Area:** The project activity instance locates in the geographic boundary for the grouped project, i.e., Indonesia, and owned by the project owner or the subsidiary of the project owner	All initial project activity instances being included in the grouped project are located within geographic boundaries of Indonesia. Hence this condition is fulfilled. Please refer to KML file for the Geographical area of project activity
3	**Baseline scenario:** All Project Activity Instances shall meet the baseline definition as defined in respective methodology	All initial project activity instances have same baseline as per methodology as detailed in subsequent sections. Hence this eligibility criteria are fulfilled. Please refer section 3.4 of this document for baseline scenario
4	**Technology type:** The project activity instances to be included in the grouped project activity is the implementation of biogas recovery, flaring and/or utilization systems in palm oil mills using covered anaerobic lagoon, and gas engine for power generation	All initial project activity instances being included in the grouped project activity are the implementation of biogas recovery, flaring and/or utilization systems in palm oil mills. The generated electricity is supplied to grid. Hence this condition is fulfilled.
5	**Additionality:** The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology	The initial project activity instances being included in grouped project activity are small scale projects and their additionality is demonstrated as discussed in section 3.5 of this document. Hence this condition is fulfilled. Please refer section 3.5 of this document for additionality.
6	**Start Date:** The start date of each project activity instance under the grouped project should not be prior	The start date of grouped project is considered on 01/09/2024 which is the date of handover of the biogas plant to

No	Eligibility Criteria	PAIs Eligibility
	to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence	the project proponent for project activity instance PT SPMN and start date of all other project activity instances is after this date. Hence this condition is fulfilled. Please refer section 1.8 of this document or commissioning certificate for start date of PAIs.

1.5 Project Proponent

Organization name	TSH Resources Berhad
Contact person	Zackaria Bin Abdulah
Title	General Manager
Address	MENARA TSH No. 8, Jalan Semantan, Damansara Heights, 50490 Kuala Lumpur, Malaysia
Telephone	603-2084-0888
Email	zackaria.abdulah@tsh.com.my

1.6 Other Entities Involved in the Project

Organization name	PT Sarana Prima Multi Niaga
Role in the project	Project Operator of Project Activity Instance 1
Contact person	Siew Chee Siong
Title	Mill Head
Address	APL Tower Lantai 11 unit 6, Jalan Letnan Jenderal Siswondo Parman, Kav.28, Grogol Petamburan, Jakarta Barat, DKI Jakarta. 11470
Telephone	+62 271-390-34255
Email	cheesiong.siew@tsh.com.my

Organization name	PT Andalas Agro Industry
Role in the project	Project Operator of Project Activity Instance 2
Contact person	Tuah Bin Mohamad
Title	Mill Head
Address	APL Tower Lantai 11 unit 6, Jalan Letnan Jenderal Siswondo Parman, Kav.28, Grogol Petamburan, Jakarta Barat, DKI Jakarta. 11470
Telephone	+6281949255727
Email	tuah.mohamad@tsh.com.my

Organization name	PT Farinda Bersaudara
Role in the project	Project Operator of Project Activity Instance 3
Contact person	Senthilkumaran A/L Gopal
Title	Mill Head
Address	APL Tower Lantai 11 unit 6, Jalan Letnan Jenderal Siswondo Parman, Kav.28, Grogol Petamburan, Jakarta Barat, DKI Jakarta. 11470
Telephone	+60196535612
Email	senthilkumaran@tsh.com.my

Organization name	PT Andalas Wahana Berjaya
Role in the project	Project Operator of Project Activity Instance 4
Contact person	P Shailendran A/L R Palpanadan
Title	Mill Head
Address	APL Tower Lantai 11 unit 6, Jalan Letnan Jenderal Siswondo Parman, Kav.28, Grogol Petamburan, Jakarta Barat, DKI Jakarta. 11470
Telephone	+6281287949537
Email	p.shailendran@tsh.com.my

1.7 Ownership

TSH Resources Berhad is the sole project owner of the project.

1.8 Project Start Date

In the year of 2024, the earliest operation start date of the four (4) project activity instances, i.e., the date on which the project began generating GHG emission reductions or removals. The expected operation start date of each project activity instance is as below:

Table 5: Project Start Date for PAIs

No	Project activity instance	Project start date
PAI 1	PT Sarana Prima Multi Niaga (SPMN)	01/09/2024
PAI 2	PT Andalas Agro Industry (AAI)	2025
PAI 3	PT Farinda Bersaudara (FDB)	2025
PAI 4	PT Andalas Wahana Berjaya (AWB)	2025

1.9 Project Crediting Period

This project adopts the renewable crediting period of 21 years, 7-year crediting period with the option to renew two more times.

The first crediting period is from 01/09/2024 to 31/08/2031 (the start and end dates included).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO_2e/year

☒ 20,000 – 100,000 tCO_2e/year

☐ 100,001 – 1,000,000 tCO_2e/year

☐ >1,000,000 tCO_2e/year

The estimated Emission Reduction for the initial project activity instances being included in the grouped project activity are as follows:

Table 6: Annual Emission Reduction Estimation for PAIs

No	Project activity instance	Estimated Annual Emission Reductions (tCO_2e)
PAI 1	PT Sarana Prima Multi Niaga (SPMN)	30,031
PAI 2	PT Andalas Agro Industry (AAI)	21,115
PAI 3	PT Farinda Bersaudara (FDB)	26,150
PAI 4	PT Andalas Wahana Berjaya (AWB)	25,877
	Total	**103,174**

As per section 3.10.1 of VCS standard Version 4.4, the projects are classified as follows:

1) Projects: Less than or equal to 300,000 tonnes of CO_2e per year
2) Large projects: Greater than 300,000 tonnes of CO_2e per year

Each PAIs being included under grouped project activity currently the emission reductions have less than 300,000 tCO_2e per year. Hence these project activity instances are classified as "Projects".

Project Scale	
Project	√
Large project	

Total emission reductions for all initial project activity instances being included in the grouped project are as below:

Year	Estimated GHG emission reductions or removals (tCO_2e)				Total (tCO_{2e})
	SPMN	FDB	AAI	AWB	
01/09/2024 – 31/12/2024	11,441	0	0	0	11,441
01/01/2025 – 31/12/2025	34,322	25,338	31,380	31,053	122,092
01/01/2026 – 31/12/2026	34,322	25,338	31,380	31,053	122,092
01/01/2027 – 31/12/2027	34,322	25,338	31,380	31,053	122,092
01/01/2028 – 31/12/2028	34,322	25,338	31,380	31,053	122,092
01/01/2029 – 31/12/2029	34,322	25,338	31,380	31,053	122,092

Year	Estimated GHG emission reductions or removals (tCO_2e)				Total (tCO_{2e})
	SPMN	FDB	AAI	AWB	
01/01/2030 – 31/12/2030	34,322	25,338	31,380	31,053	122,092
01/01/2031 – 31/08/2031	22,881	16,892	20,920	20,702	81,395
Total estimated ERs	**240,251**	**168,919**	**209,198**	**207,020**	**825,388**
Total number of crediting years	7	7	7	7	
Average annual ERs	**30,031**	**21,115**	**26,150**	**25,877**	**103,174**

The above annual estimation for all initial PAIs is based on estimated amount of POME, COD amount obtained from the 10 days baseline campaign, and total estimated gas engine capacity of all project activity instances. Thus, actual emission reductions will be varied based on actual monitoring data and commissioning date of the PAIs.

1.11 Description of the Project Activity

Description of the Project Activity - Grouped Project

The Grouped Project will consist of multiple PAIs for the implementation of biogas recovery, flaring and/or utilization systems from industrial wastewater in Indonesia. The Grouped Project will reduce greenhouse gas (GHG) emissions from sources of industrial wastewater by capturing the biogas that would otherwise have been generated and released to the atmosphere from the baseline wastewater treatment systems.

Description of the Project Activity – Project Activity Instances

Baseline Scenarios

PAI1 – PT Sarana Prima Multi Niaga (SPMN)

SPMN Palm Oil Mill is an established palm oil mill that is processing approximately 234,180 mt/yr of Fresh Fruit Bunches (FFB). The processing of FFB creates a significant volume of wastewater that requires treatment, due to a high level of organic matter, as measured by the wastewater's Chemical Oxygen Demand (COD). The existing wastewater treatment system includes a series of ponds (anaerobic and aerobic). The breakdown of COD in an anaerobic environment result in the emission of methane into the atmosphere.

PAI2 – PT Andalas Agro Industry (AAI)

AAI Palm Oil Mill is an established palm oil mill that is processing approximately 149,430 mt/yr of Fresh Fruit Bunches (FFB). The processing of FFB creates a significant volume of wastewater that requires treatment, due to a high level of organic matter, as measured by the wastewater's Chemical Oxygen Demand (COD). The existing wastewater treatment system includes a series of ponds (anaerobic and aerobic). The breakdown of COD in an anaerobic environment result in the emission of methane into the atmosphere.

PAI3 – PT Farinda Bersaudara (FDB)

FDB Palm Oil Mill is an established palm oil mill that is processing approximately 298,803 mt/yr of Fresh Fruit Bunches (FFB). The processing of FFB creates a significant volume of wastewater that requires treatment, due to a high level of organic matter, as measured by the wastewater's Chemical Oxygen Demand (COD). The existing wastewater treatment system includes a series of ponds (anaerobic and aerobic). The breakdown of COD in an anaerobic environment result in the emission of methane into the atmosphere.

PAI4 – PT Andalas Wahana Berjaya (AWB)

AWB Palm Oil Mill is an established palm oil mill that is processing approximately 134,617 mt/yr of Fresh Fruit Bunches (FFB). The processing of FFB creates a significant volume of wastewater that requires treatment, due to a high level of organic matter, as measured by the wastewater's Chemical Oxygen Demand (COD). The existing wastewater treatment system includes a series of ponds (anaerobic and aerobic). The breakdown of COD in an anaerobic environment result in the emission of methane into the atmosphere.

Project Scenarios

The systems to be implemented for the PAIs is tabulated below:

Table 7: Technologies for PAIs

PAIs	Brief Description
PAI1 – PT Sarana Prima Multi Niaga (SPMN)	• Biogas digester • Gas engines • Open flare
PAI2 – PT Andalas Agro Industry (AAI)	• Biogas digester • Gas engines

PAIs	Brief Description
	• Open flare
PAI3 – PT Farinda Bersaudara (FDB)	• Biogas digester • Open flare
PAI4 - PT Andalas Wahana Berjaya (AWB)	• Biogas digester • Gas engines • Open flare

Table 8: Technologies for PAIs

Project Components	Brief Description
Anaerobic Digestion Plant	The Biogas capturing plant or Anaerobic digestion plant is simple and smaller footprint, effective and reliable technology to capture biogas that contain methane, produced during anaerobic process. Both technologies will be using HDPE layer to cover and seal the top of the POME ponds, and effectively capture biogas with more than 55% methane content produced from the system.
Biogas treatment system	This includes a pair of scrubbers and a dehumidifier and chiller. Biological scrubbers are used to oxidize hydrogen sulphide (H_2S) to sulphate. The result of the scrubbing process is clean biogas with H_2S content reduced to less than 250ppm. The biogas is passed from the scrubber to a dehumidifier/chiller system to reduce the water content in the biogas, from 75-90% to 40%, and to cool the biogas to the point it can be used in the biogas engine
Flare	Open flare will be installed at the plant. The capacity of the flare is around 1,200Nm^3/hr. An open flare is also installed, although it is not expected to be needed due to the storage capacity of the biogas membrane (emissions from the flare are not expected to be material).
Gas engine	Biogas engine will be installed at the Biogas Plant.

Project Components	Brief Description
PLC and SCADA	State-of-art PLC and SCADA will be used to operate and monitor the project.
Aerobic ponds for post-treatment	The remaining liquid will be further treated in aerobic ponds equipped with surface aerators as well as a final polishing plant.

1.12 Project Location

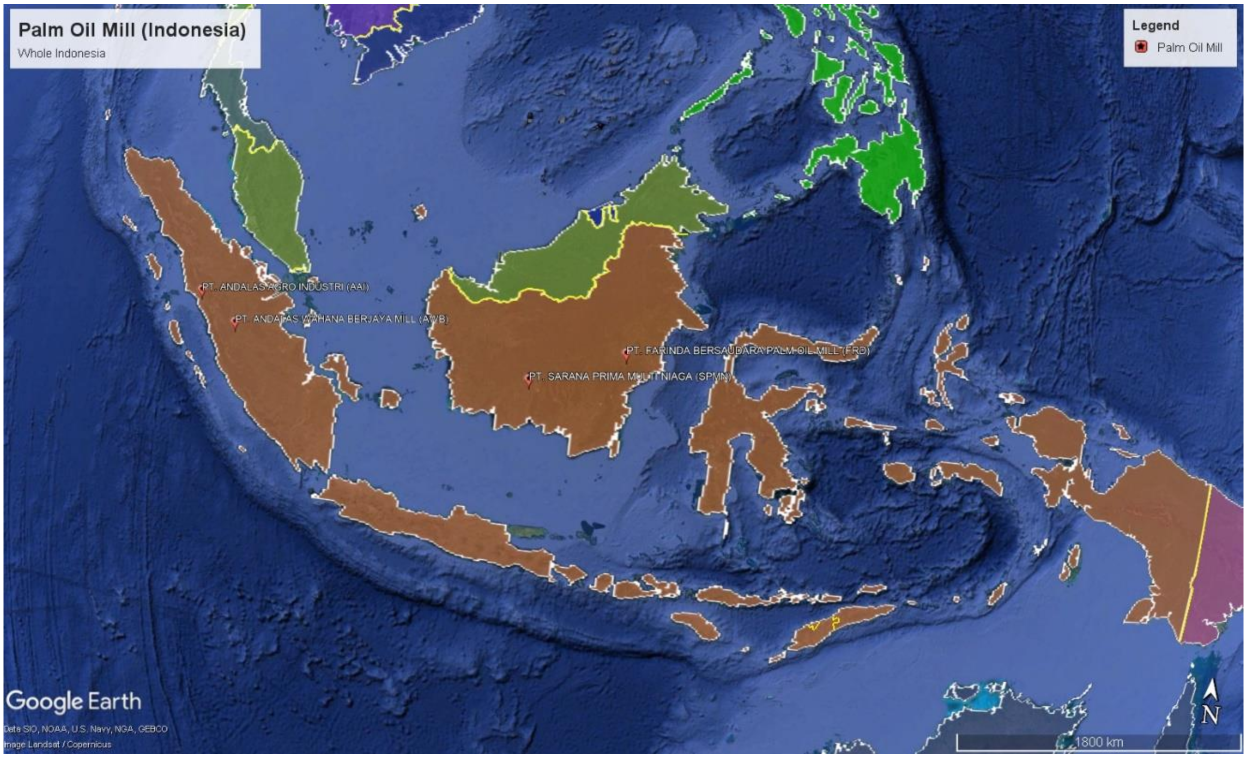
Geographic area for the Grouped Project

Project Activity Instances within the border of Indonesia are allowed to be added to and included in this Grouped Project. Geodetic coordinates are provided as a KML file.

Geographic area for PAIs

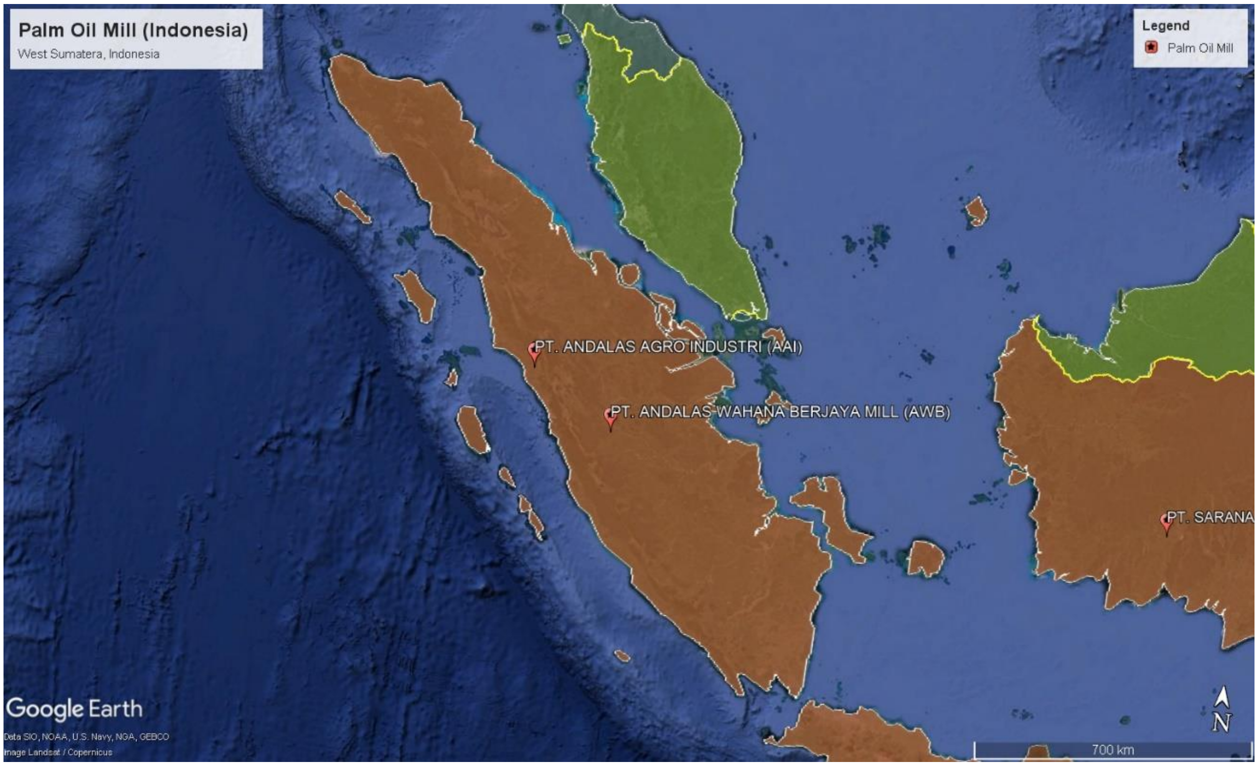
The geographical location of each project activity instance under the grouped project is provided as below:

No	Project activity instance	Geographical Coordinates
PAI1	PT Sarana Prima Multi Niaga (SPMN)	S 1°57'11.50" E 12°55'41.08"
PAI2	PT Andalas Agro Industry (AAI)	S 00 03' 21" E 99 58' 21"
PAI3	PT Farinda Bersaudara (FDB)	S 0°41'43.23" E 16°15'41.98"
PAI4	PT Andalas Wahana Berjaya (AWB)	S 1° 7'12.26" E 101°33'5.48"



Palm Oil Mill (Indonesia)
Whole Indonesia
Legend
Palm Oil Mill
PT. ANDALAS AGRO INDUSTRI (AAI)
PT. ANDALAS WAHANA BERJAYA MILL (AWB)
PT. FARINDA BERSAUDARA PALM OIL MILL (FRD)
PT. SARANA PRIMA MULTI NIAGA (SPMN)
Google Earth
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Image Landsat / Copernicus
1800 km
N

Palm Oil Mill (Indonesia)
Central & East Kalimantan, Indonesia
Legend
Palm Oil Mill
PT. FARINDA BERSAUDARA PALM OIL MILL (FRD)
PT. SARANA PRIMA MULTI NIAGA (SPMN)
Google Earth
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Image Landsat / Copernicus
400 km
N



1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of each project activity instance is: The wastewater (POME) was left at uncovered open lagoon and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. The conditions existing prior to the project initiation are the same as the baseline scenario, please refer to Section 3.4 (Baseline Scenario) for further details.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Indonesia relevant laws and regulations. Mainly includes the following:

Table 9: Laws and Regulations for Indonesia

Laws	Description
Laws related to Electricity Generation	
Law No. 30/2009	Law regulating the general guidance of Indonesian electricity.
Government Regulation No. 79/2014	The government regulation related to national energy policy which determines a target share of new and renewable energy in primary energy of 23% by 2025 and

Laws	Description
	31% by 2050.
Laws related to Wastewater	
Ministry of Environment Decree No. 5/2014	Indonesia applicable regulations on wastewater treatment are Government related to Water Quality Management and Water pollution Control and the Ministry of Environment Decree No. 5/2014 which sets maximum levels (mg/L) and maximum pollution load of wastewater discharged into the environment. The permits show that the mill has been complying with rules and regulation in terms of discharge level of biochemical oxygen demand (BOD) set by the Department of Environment Indonesia.
Laws related to Biogas	
None	There is no regulation in Indonesia that requires the management of biogas from the project site and/or its combustion for power generation. There is no obligation to recover and/or use the biogas.
Laws related to requirement to conduct EIA	
Ministry of Environment Decree No. 11/2006	Types of business and activity plans that must be completed with an analysis of environmental impacts

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered and is not seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project emission reductions activities have not been included in an emissions trading program or any other mechanism.

1.16.2 Other Forms of Environmental Credit

The Project has not sought or received any other form of GHG-related environmental credit.

Supply Chain (Scope 3) Emissions

Have the owner(s) or retailer(s) of the impacted goods and services2 posted a public statement saying, "VCUs may be issued for the greenhouse gas emission reductions and removals associated with [organization name(s)] [name of good or service]" since the project's start date?

☐ Yes ☒ No

Explain your response.

Has the project proponent posted a public statement saying, "VCUs may be issued for the greenhouse gas emission reductions and removals associated with [name of good or service][describe the region or location, including organization name(s), where practicable]."

☐ Yes ☒ No

Explain your response.

Have the producer(s) or retailer(s) of the impacted good or service been notified of the project and the potential risk of Scope 3 emissions double claiming via email?

☐ Yes ☒ No

Explain your response.

In all other cases, demonstrate that a public statement(s) by the owner(s) or retailer(s) of the impacted good*(s)* or service*(s)* or project proponent (as applicable) has been made

[2] Impacted goods and services are all goods and services directly impacted by the technologies and measures specified as project activities in the project description. Please see the VCS Program document *VCS Program Definitions* for additional information.

throughout the project crediting period. Where applicable, also demonstrate that the impacted good or service's producer(s) or retailer(s) have been notified of the project and the potential risk of Scope 3 emissions double claiming via email. Evidence of the public statement(s) and email(s) must be provided in this report or attached as an appendix.

1.17 Sustainable Development Contributions

The project activity aims to introduce new biogas digester or covering of anaerobic lagoon to replace the current open anaerobic lagoons. With these new facilities, methane emissions from the existing open lagoons are significantly avoided. Biogas generated in the covered lagoon is recovered as renewable energy. The project activity contributes to sustainable development in the following aspects:

(i) **Social**

- Creates employment and business opportunities for the locals;
- Provides the local community (workers on-site) a sustainable, clean source of energy; and
- Avoidance of methane emissions from the wastewater treatment system which indirectly benefits the health of workers and adjacent residents.

(ii) **Economic**

- Attracts foreign investment and cash flow to Indonesia from Certified Emission Reduction (CER) trading;
- Reduce land required for POME treatment;
- Technological innovation and improvement by promoting cleaner technology and reducing emissions of greenhouse gasses (GHGs); and
- Promotes a green image for Indonesian palm oil industry.

(iii) **Environment**

- Reduces air pollutants (GHG) that contribute to climate change and global warming;
- Improves local air quality (reduction in methane from lagoons); and
- Promotes waste recycling and contributes to resource conservation.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable. In accordance with AMS III.H, Ver. 19, no leakage emissions are considered.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

None.

2 SAFEGUARDS

2.1 No Net Harm

There is no negative environmental or socio-economic impact. The project activity involves the implementation of anaerobic wastewater management system and the combustion of biogas as per VCS definitions and requirements. The project follows the local regulation to maintain and prevent environmental impact.

The infrastructure for export of electricity is using existing electricity towers that do not give rise to concerns to villagers.

2.2 Local Stakeholder Consultation

To be developed

2.3 Environmental Impact

Every additional PAI to be added to the Grouped Project is required to summarise any environmental impact assessments carried out with respect to the PAI, where applicable.

PAI1, PAI2, PAI3 and PAI4

There is no legal requirement to conduct new Environmental Impact Assessment according to Ministry of Environment Decree No. 11/2006, as the nature of the PAIs are *recognized by the national government as unlikely to result in raising environmental impacts to the environment.*

The development of the PAI's is a net positive, owing to the reduction of carbon emissions, odour and waste emissions. Though there is no legal requirement for Environmental Impact Assessment, relevant documents have been submitted for review of monitoring and management of environmental impacts (UKL/UPL).

2.4 Public Comments

To be developed

2.5 AFOLU-Specific Safeguards

Not applicable as this is a non-AFOLU project

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3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The following methodologies are applicable to the Grouped Project:

(1) Methane avoidance component:

AMS-III.H: Methane recovery in wastewater treatment --- Version 19.0;

The following tools are applied:

- Project emissions from flaring, version 04.0;
- Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion, version 03.0
- Project and leakage emissions from anaerobic digesters, version 01.0.0, and
- Positive lists of technologies, version 04.0

(2) Electricity generation component (export to the grid):

AMS-I.D: Grid connected renewable electricity generation --- Version 18.0

The following tools are also applied:

- Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion, version 03.0

(3) Electricity generation component (for own captive and mini grid use):

AMS-I.F: Renewable electricity generation for captive use and mini-grid --- Version 05.0

The following tools are also applied:

- Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
- Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion, version 03.0

Note that PAIs within the Grouped Project may not require the use of all the above approved methodologies and methodological tools. Relevance and/or applicability should be assessed on a case by-case basis.

3.2 Applicability of Methodology

The project includes small-scale activities for which methodologies of type AMS-III.H and AMS-I.D and/or AMS-I.F are applicable.

Table 9: Applicability conditions for AMS-III.H. version 19.0

No.	Applicability Criteria	Project activity instance	Applicability
1.	The methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; (b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; (e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a	The PAIs to be implemented under this Grouped Project is to recover the biogas from wastewater treatment system which would have otherwise been vented to the atmosphere. Option (f) as in the methodology will be applied to the PAIs.	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
	sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).		
2.	In cases where baseline system is anaerobic lagoon, the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15°C, at least during part of the year on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	The baseline system is anaerobic lagoon of which: (a) The lagoons are deeper than 2 meters and without aeration[3]. The value is obtained from an engineering design document; (b) The average ambient temperature around Indonesia is reported to be 30°C[4] which is way above 15°C; (c) The interval of the removal of sludge is more than 30 days.	Applicable
3.	The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case, additional guidance provided in Annex 1 shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case, additional guidance	The PAIs involve facilities to utilize biogas generated for renewable energy, and excess gas is burned via flaring. PAIs will have an option to utilize the biogas recovered for energy, upgrading or transportation and/or	Applicable

[3] The values are obtained from an engineering design document provided by respective PAIs.

[4] https://weather-and-climate.com/average-monthly-Rainfall-Temperature-Sunshine-in-Indonesia

No.	Applicability Criteria	Project activity instance	Applicability
	provided in Annex 1 shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users. (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	bottled for use and/or sold to the grid, depending on each site specification and requirement.	
4.	If the recovered biogas is used for project activities covered under paragraph 3 (a), that component of the project activity can use a corresponding methodology under Type I.	The methane captured will be utilized for renewable electricity generation of which the approved methodologies AMS-I.D. and/or AMS-I.F will be used. PAIs will have different project activities, for the electricity generation component of the project activity (AMS-I.D), and upgrading of biogas to be used for transportation applications (AMS-I.F).	Applicable
5.	For project activities covered under paragraph 3 (b), if bottles with upgraded biogas are sold outside the project boundary, the end use of the biogas shall be ensured via a	There will be upgrading and distribution of bottled biogas for the PAIs involved in this grouped project.	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
	contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period, CO_2 emissions avoided by the displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. AMS-I.C *Thermal energy production with or without electricity*.	Contracts are available to demonstrate agreement on claiming of emission reduction only for project proponent, TSH Resources Berhad.	
6.	For project activities covered under paragraph 3 (c) (i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	The PAIs will have option to upgrade and injecting gas to the grid. The Barito grid (Central Kalimantan) of Indonesia is located within the host country and not connected to any grid outside of host country[5]. PAIs located in Sumatera Island connects with Java, Madura and Bali region, and also not connected to any grid	Applicable

[5] https://www.beritadaerah.co.id/2022/03/18/proyeksi-kebutuhan-tenaga-listrik-kalimantan-tengah/

No.	Applicability Criteria	Project activity instance	Applicability
		outside of host country[6].	
7.	For project activities covered under paragraph 3 (c) (ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	There will be no activities of transportation of natural gas distribution via a dedicated piped network to a group of end user(s), for the PAIs.	Not applicable.
8.	In particular, for the case of 3 (b) and (c) (iii), the physical leakage during storage and transportation of upgraded biogas as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of AMS-III.H *Methane recovery in wastewater treatment* shall be followed in this regard.	Captured methane will be treated for transportation applications, and also bottled and sent to end users. Leakage during transportation and storage as well as from transportation of vehicles are considered for the project activity instance under grouped project activity	Applicable
9.	For project activities covered under paragraph 3 (b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these exist) or in the absence of national regulations, a minimum of 96% (by	The project activity instance under grouped project activity will be upgrading of biogas, and bottling as well as distribution of the biogas. There has been	Applicable

[6] https://www.energytransitionpartnership.org/uploads/2022/09/The-Importance-of-Upgrading-Control-Center-to-Integrate-More-Variable-Renewable-Energy-in-PLN-Jamali-Control-Center-System.pdf

No.	Applicability Criteria	Project activity instance	Applicability
	volume).	no national regulations in-force at the time of writing PD. Ministry of Energy and Mineral Resources has announced its it is currently revising related regulations, on Biomethane licensing[7].	
10.	If the recovered biogas is utilized for the production of hydrogen (project activities covered under paragraph 3 (d)), that component of the project activity shall use the corresponding methodology AMS-III.O *Hydrogen production using methane extracted from biogas*.	There will be no production of hydrogen, for the project activity instance under grouped project activity	Not applicable
11.	If the recovered biogas is used for project activities covered under paragraph 3 (e), that component of the project activity shall use the corresponding methodology AMS-III.AQ *Introduction of Bio-CNG in road transportation*.	The recovered biogas for the project activity instance under grouped project activity, will not use for road transportation applications after upgrading, for the PAIs.	Not Applicable
12.	New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system	The project activity instance under grouped project activity is upgrading of existing facilities, and	Not applicable

7 https://ebtke.esdm.go.id/post/2023/03/13/3446/dukung.pemanfaatan.bioenergi.kini.resmi.terbit.perizinan.biogas.sebagai.bahan.bakar.lain

No.	Applicability Criteria	Project activity instance	Applicability
	compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the *General guidelines to SSC CDM methodologies*. In addition, the requirements for demonstrating the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	thus is not a greenfield project.	
13.	The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the Project Design Document (PDD).	The location of each PAIs will be identified by specific reference code, address, map and GPS coordinates. The unique location of the wastewater generating source will be defined and described in Section 1.12. of Project Description (PD), and also shown in the baseline and project activity diagram	Applicable
14.	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 $ktCO_2$ equivalent annually from all Type III components of the project activity.	The project activity instance under grouped project activity will result in less than or equal to $60ktCO_2$ equivalent annually from all Type III, as demonstrated in section 1.10 of PD.	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
		The estimated average emission reductions from Type III component (methane avoidance) over the crediting period for all the PAIs is lower than the 60 ktCO_2e annual threshold limit for Type III component (methane avoidance).	

Table 10: Applicability conditions for AMS-I.D. version 18.0

No.	Applicability Criteria	Project activity instance	Applicability
1.	This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	The project activity instance under grouped project activity utilizes the captured biogas from Palm Oil Mill Effluent for electricity generation for own mill use, and supply electricity to workers housing estate.	Applicable
2.	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s);	The project activity instances under grouped project activity involves retrofitting existing facility at site for wastewater treatment.	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
	(d) Involve a rehabilitation of (an) existing plant(s)/ unit(s); or (e) Involve a replacement of (an) existing plant(s).	There is no capacity addition of existing plant, rehabilitation, or replacement of existing plant(s).	
3.	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than $4W/m^2$; or (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than $4W/m^2$.	The project activity instances under grouped project activity are not hydro power plant project.	Not applicable
4.	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the	The project activity instances under grouped project activity have renewable component of less than 15MW capacity.	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
	limit of 15MW.		
5.	Combined heat and power (co-generation) systems are not eligible under this category.	The project activity instances under grouped project activity are not a co-generation project.	Not applicable
6.	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15MW and should be physically distinct from the existing units.	The project activity instances under grouped project activity's implementation are not at an existing renewable power generation and does not involve any capacity addition of renewable energy generation unit.	Not applicable
7.	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/ unit shall not exceed the limit of 15MW.	The project activity instances under grouped project activity involve retrofit of existing facilities and does not exceed the limit of 15MW.	Applicable
8	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to the grid then the baseline for the	The project activity instances under grouped project activity are wastewater treatment project, and eligible under Type III category. In the absence of the project,	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
	electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as 'AMS-I.C: Thermal energy production with or without electricity' shall be explored.	the existing facility will continue to operate.	
9	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool 'Project emissions from cultivation of biomass' shall apply.	The project activity instances under grouped project activity are not using biomass sourced from dedicated plantations.	Not applicable.

Table 10: Applicability conditions for AMS-I.F. version 05.0

No.	Applicability Criteria	Project activity instance	Applicability
1.	This methodology comprises renewable energy generation units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass that supply electricity to user(s). The project activity will displace electricity from an electricity distribution system that is or would have been supplied by at least one fossil fuel fired generating unit, i.e. in the absence of the project activity, the users would have been supplied electricity from one or more sources listed below: (a) A national or regional grid	The project activity instances under grouped project activity are renewable energy generating unit using methane gas, which displaces fossil fuel fired electricity using diesel engine that is supplied to user(s), i.e. carbon intensive mini-grid.	Applicable

No.	Applicability Criteria	Project activity instance	Applicability
	(grid hereafter); or (b) A fossil fuel fired captive power plant; (c) A carbon intensive mini-grid.		
2	The methodology is applicable for project activities that: (a) install a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant); (b) Involve a capacity addition; (c) Involve a retrofit of (an) existing plant(s); or (d) involve a replacement of (an) existing plant(s).	The project activity instances under group project activity will be installed at locations which do not have renewable power plant operating prior to implementation of the project, by way of retrofitting of existing facility.	Applicable
3	Illustration of respective situations under which each of the methodology ("AMS-I.D: Grid connected renewable electricity generation", "AMS-I.F: Renewable electricity generation for captive use and mini-grid" and "AMS-1.A: Electricity generation by the user") applies is included in Table 2.	The project activity instances under group project activity apply AMS-I.F, "Renewable electricity generation for captive use and mini-grid".	Applicable
4	In case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity the units added by the project should be lower than 15MW and should be physically distinct from the existing units.	The project activity instances under group project activity do not involve any capacity addition to renewable energy generation units to an existing renewable power generation facility.	Not applicable

No.	Applicability Criteria	Project activity instance	Applicability
5	In the case of retrofit replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement unit shall not exceed the limit of 15MW.	The project activity instances under group project activity involved retrofitting of existing facility, but with capacity of lower than 15MW.	Applicable
6	If the unit added has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the unit added co-fires fuel, the capacity of the entire unit shall not exceed the limit of 15MW.	The project activity instances under group project activity's project addition involve only renewable components, of lower than 15MW capacity.	Applicable
7	Combined heat and power (co-generation) systems are not eligible under this category.	The project activity instances under group project activity do not involve combined heat and power generation systems.	Not applicable
8	Hydropower plants with reservoirs that satisfy at least one of the following conditions are eligible to apply for this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of the reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of	The project activity instances under group project activity are not hydropower plant project.	Not applicable

No.	Applicability Criteria	Project activity instance	Applicability
	the reservoir is increased and the power density of the project activity as per definitions given in the project emissions section, is greater than $4W/m^2$; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than $4W/m^2$.		
9	If electricity and/or steam/heat produced by the project activity is delivered to a third party, i.e. another facility or facilities within the project boundary, a contract between the supplier and consumer(s) of the energy will have to be entered that ensures that there is no double counting of emission reductions.	The project activity instances under group project activity may export the electricity generated to a third party. Relevant documents will be produced to ensure no double counting of emission reductions on both parties.	Applicable
10	In the case of project activities utilizes biomass, the 'TOOL16: Project and leakage emissions from biomass" shall be applied to determine the relevant project emissions from the cultivation of biomass and the utilization of biomass or biomass residues.	The project activity instances under group project activity do not utilize biomass, and thus cultivation and utilization of biomass/ biomass residues will not be applied.	Not applicable

3.3 Project Boundary

The project boundary is the physical, geographical site where the wastewater and sludge treatment take place, covering all facilities affected by the project activity including sites where processing, transportation and application of disposal of waste products as well as the biogas takes place, including:

- The project power plant and all power plants connected physically to the electricity system that the project power plant is connected to,
- Distributed users who are connected via mini/isolated grid (s) extending to power plant and all power plants connected physically to the electricity system which the project power plant is connected, and
- Upgrade and compression installation plant for bottling of captured biogas, and transportation to end user.

Summary of greenhouse gases and sources included in and excluded from the project boundary are summarised below:

Table 11: Emission sources included or excluded for AMS-III.H in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions on account of electricity or fossil fuel used	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from baseline wastewater treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from baseline sludge treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
	river/lake/sea			
	Methane emissions from the decay of the final sludge generated by the baseline treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from injection of upgraded biogas into a natural gas distribution grid	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO_2 emissions from electricity and fuel used by the project facilities	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

	Source	Gas	Included?	Justification/Explanation
	Methane emissions from the decay of the final sludge generated by the project activity treatment systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane fugitive emissions due to inefficiencies in capture systems	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions due to incomplete flaring	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Emissions from electricity and fuel used by upgrading facilities	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Methane emissions from the discharge of the upgrading equipment	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
	Fugitive methane emissions from leaks in compression equipment	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification

Source		Gas	Included?	Justification/Explanation
		Other	No	Excluded for simplification
	Emissions on account of vent gases from upgrading equipment	CO_2	No	Excluded for simplification
		CH_4	Yes	Main emission source
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 12: Emission sources included or excluded for AMS- I.D in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in power plants that are displaced due to the project activity	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO_2 emissions from on-site consumption of fossil fuel due to project activity	CO_2	Yes/No	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification

Table 13: Emission sources included or excluded for AMS-I.F in the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from electricity displaced with the electricity produced by the renewable generating unit and emission factor	CO_2	Yes	Main emission source
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	NIL	CO_2	No	NIL
		CH_4	No	NIL
		N_2O	No	NIL
		Other	No	NIL

3.4 Baseline Scenario

According to AMS-III.H version 19.0, the baseline scenario involves a baseline system with anaerobic wastewater treatment system without biogas recovery, which is the case the baseline scenario is determined to be the continuation of the existing open lagoon-based treatment of POME without methane recovery as commonly practiced in the palm oil processing industries because it only requires small investment and maintenance cost where the treatment system only consume less energy for pumping and mixing.

According to AMS-I.D version 18.0, the baseline scenario for greenfield power plant, as proposed project is not connected to the grid and that there is no renewable energy power plant operated prior to the implementation of the project activity.

According to AMS-I.F version 05.0, the baseline scenario is electricity would have been supplied from an electricity distribution system that is or would have been supplied by at least one fossil fuel fired generating unit.

Therefore, as a result of the combination of baseline analysis for all the above methodologies, it is concluded that the baseline scenario of the proposed project is, in the absence of the project activity:

- Wastewater would have been treated in anaerobic open lagoons.
- Electricity for mini-grid within the project activity would have otherwise been generated by the operation of fossil-fuel fired power plants and/or fossil fuel fired generator.
- Grid electricity would have been generated by operation of grid connected power plants and by addition of new generation sources into the grid.

PAI1 – PT Sarana Prima Multi Niaga (SPMN)

The situation prior to the implementation of the is represented as follows:

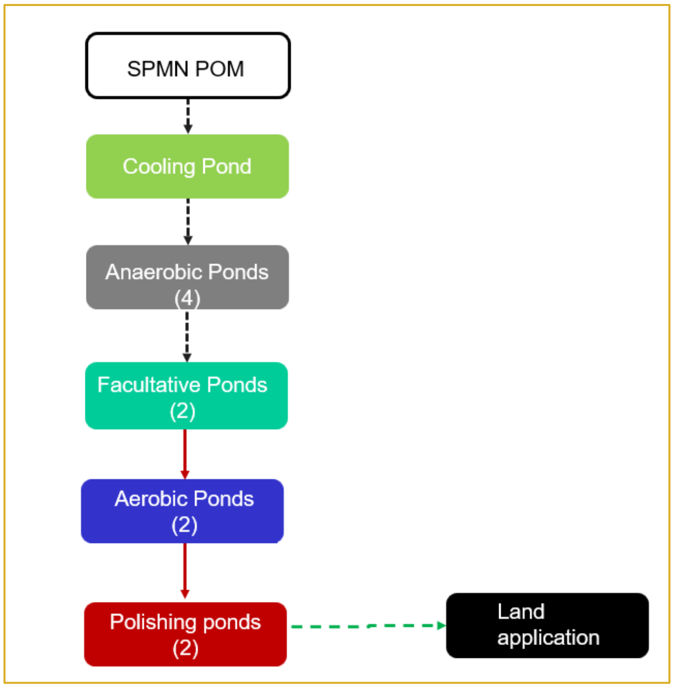

Figure 1: PAI1 - Sarana Prima Multi Niaga (SPMN) Baseline Scenario

PAI2 – PT Andalas Agro Industry (AAI)

The situation prior to the implementation of the is represented as follows:

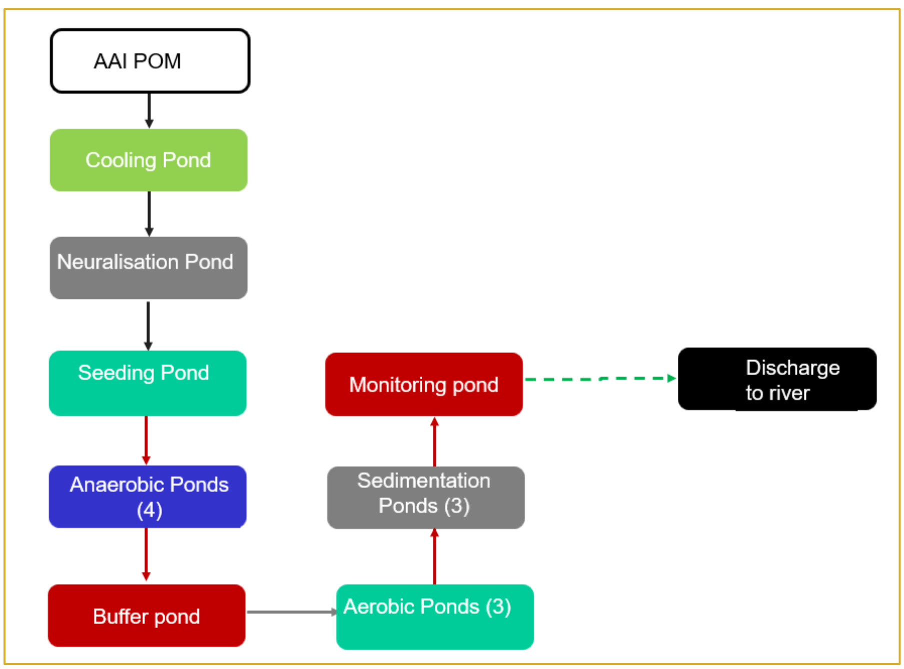

Figure 2: PAI2 – Andalas Agro Industry (AAI) Baseline Scenario

PAI3 – PT Farinda Bersaudara (FDB)

The situation prior to the implementation of the is represented as follows:

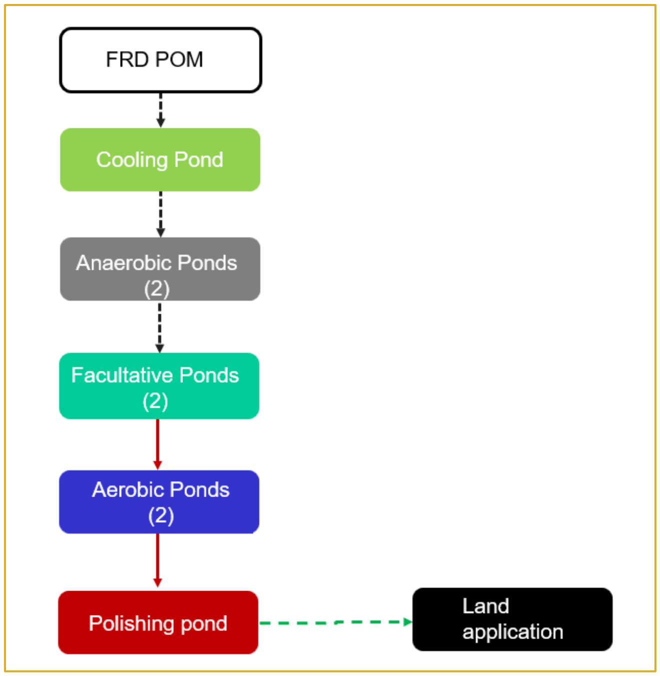

Figure 3: PAI3 *Farinda Bersaudara (FDB)* Baseline Scenario

PAI4 – PT Andalas Wahana Berjaya (AWB)

The situation prior to the implementation of the is represented as follows:

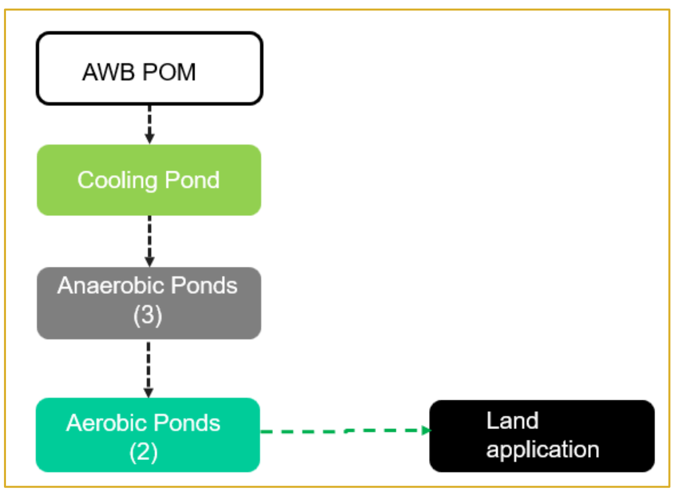

Figure 4: PAI4 *Andalas Wahana Berjaya (AWB)* Baseline Scenario

Project Scenario

The detailed systems to be implemented for the PAIs is tabulated below:

Table 14: Technologies for PAIs

PAIs	Biogas Digester	Gas Engines	Open Flare

PAIs	Biogas Digester	Gas Engines	Open Flare
PAI1 – PT Sarana Prima Multi Niaga (SPMN)	√	√	√
PAI2 – PT Andalas Agro Industry (AAI)	√	√	√
PAI3 – PT Farinda Bersaudara (FDB)	√		√
PAI4 - PT Andalas Wahana Berjaya (AWB)	√	√	√

The four (4) PAIs will introduce biogas digester system, a biogas treatment system, and a power generation system. The technologies description is mentioned in section above.

3.5 Additionality

As per tools Positive lists of technologies - Version 04.0, A positive list of technology and project activity types that are defined as automatically additional if the PAIs apply such technologies and meet specified conditions. The positive list of technology including the methane recovery in wastewater treatment where the specific conditions to be met is tabulated below:

Table 15: Additionality for PAIs

Requirements for Additionality	Project Scenario
(a) The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation	*Yes* The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the regulations of Indonesia.
(b) There is no regulation in the host country that requires the management of biogas from domestic, industrial and	Yes There is no regulation in Indonesia that requires the management of biogas from a site such as the project site.

Requirements for Additionality	Project Scenario
agricultural sites	
(c) There is no capacity increase in the wastewater treatment system	Yes There has been no capacity increase in the wastewater treatment system
(d) No other alternative economic activity is expected to be undertaken on the land of the existing lagoon	Yes No other alternative economic activity is expected to be undertaken on the land of the existing lagoon.
(e) The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW.	Yes The biogas is used to generate electricity in a power plant and the total nameplate capacity is below 5MW.

PAI1, PAI2, PAI3 and PAI4 all meet the five (5) above conditions. Thus, all the PAIs are automatically additional.

3.6 Methodology Deviations

No methodology deviations

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline Emissions for the Grouped Project

AMS III H, Version 19.0

According to the approved methodology AMS III H, Version 19.0, the baseline emissions for instances in the Grouped Project are determined according to Equations 1 to 7 as follows:

1. Baseline emissions for the systems affected by the project activity may consist of:
   (a) Emissions on account of electricity or fossil fuel used ($BE_{power,y}$);
   (b) Methane emissions from baseline wastewater treatment systems ($BE_{ww,treatment,y}$);
   (c) Methane emissions from baseline sludge treatment systems ($BE_{s,treatment,y}$);
   (d) Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea ($BE_{ww,discharge,y}$);
   (e) Methane emissions from the decay of the final sludge generated by the baseline treatment systems ($BE_{s,final,y}$).

$$BE_y = \{BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y}\} \quad \text{Equation (1)}$$

Where:

Parameter	Description	Unit
BE_y	Baseline emissions in year y	tCO_2e
$BE_{power,y}$	Emissions on account of electricity or fossil fuel used	tCO_2e
$BE_{ww,treatment,y}$	Methane emissions from baseline wastewater treatment systems	tCO_2e
$BE_{s,treatment,y}$	Methane emissions from baseline sludge treatment systems	tCO_2e
$BE_{ww,discharge,y}$	Methane emissions on account of inefficiencies in	tCO_2e

Parameter	Description	Unit
	the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/lake/sea	
$BE_{s,final,y}$	Methane emissions from the decay of the final sludge generated by the baseline treatment systems	tCO_2e

2. Baseline emissions from electricity and fossil fuel consumption ($BE_{power,y}$) are determined as per the procedures described in the "Tool to calculate baseline, project and/or leakage emissions from electricity consumption" and "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion", respectively. The energy consumption shall include all equipment/devices in the baseline wastewater and sludge treatment facility. If recovered biogas in the baseline is used to power auxiliary equipment it should be taken into account, accordingly, using zero as its emission factor

3. Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant:

$$BE_{ww,treatment,y} = \sum_{i}\left(Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}\right) \times B_{o,ww} \times UF_{BL} \times GWP_{CH4} \qquad \text{Equation (2)}$$

Where:

Parameter	Description	Unit
$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system i in year y (m^3). For ex ante estimation, forecasted wastewater generation volume or the designed capacity of the wastewater treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated wastewater	m^3
$COD_{inflow,i,y}$	Chemical oxygen demand of the wastewater inflow to the baseline treatment system *i* in year *y* (t/m^3). Average value may be used through sampling with the confidence/precision level 90/10	t/m^3

Parameter	Description	Unit
$\eta_{COD,BL,i}$	COD removal efficiency of the baseline treatment system *i*, determined as per the paragraphs 35, 36 or 37 of AMS-III.H.	%
$MCF_{ww,treatment,BL,i}$	Methane correction factor for baseline wastewater treatment systems *i* (MCF values as per Table 2 of AMS-III.H.)	-
i	Index for baseline wastewater treatment system	-
$B_{o,ww}$	Methane producing capacity of the wastewater (IPCC value of 0.25 kg CH_4/kg COD)[8]	kg CH_4/kg COD
UF_{BL}	Model correction factor to account for model uncertainties (0.89)[9]	-
GWP_{CH4}	Global Warming Potential for methane	tCO_{2e}/tCH_4

4. Methane emissions from the baseline sludge treatment systems affected by the project activity are determined using the methane generation potential of the sludge treatment systems:

$$BE_{treatment,s,y} = \sum_j S_{j,BL,y} \times MCF_{s,treatment,BL,j} \times DOC_s \times UF_{BL} \times DOC_F \times F \times 16/12 \times GWP_{CH4} \qquad \text{Equation (3)}$$

Where:

Parameter	Description	Unit

[8] Project activities may use the default value of 0.6 kg CH_4/kg BOD, if the parameter $BOD_{5,20}$ is used to determine the organic content of the wastewater. In this case, baseline and project emissions calculations shall use BOD instead of COD in the equations, and the monitoring of the project activity shall be based in direct measurements of $BOD_{5,20}$, i.e. the estimation of BOD values based on COD measurements is not allowed.

[9] Reference: FCCC/SBSTA/2003/10/Add.2, page 25.

Parameter	Description	Unit
$S_{j,BL,y}$	Amount of dry matter in the sludge that would have been treated by the sludge treatment system j in the baseline scenario (t). For ex ante estimation, forecasted sludge generation volume or the designed capacity of the sludge treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated sludge	t
j	Index for baseline sludge treatment system	-
DOC_s	Degradable organic content of the untreated sludge generated in the year y (fraction, dry basis). Default values of 0.5 for domestic sludge and 0.257 for industrial sludge[10] shall be used	-
$MCF_{s,treatment,BL,j}$	Methane correction factor for the baseline sludge treatment system j (*MCF* values as per Table 2 of AMS-III.H.)	-
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-
DOC_F	Fraction of DOC dissimilated to biogas (IPCC default value of 0.5)	-
F	Fraction of CH_4 in biogas (IPCC default of 0.5)	-

5. If the sludge is composted, the following equation shall be applied:

$$BE_{s,treatment,y} = \sum_{j} S_{j,BL,y} \times EF_{composting} \times GWP_{CH4} \quad \text{Equation (4)}$$

10 The IPCC default values of 0.05 for domestic sludge (wet basis, considering a default dry matter content of 10 per cent) or 0.09 for industrial sludge (wet basis, assuming dry matter content of 35 per cent), were corrected for dry basis.

Where:

Parameter	Description	Unit
$EF_{composting}$	Emission factor for composting organic waste (tCH_4/t waste treated). Emission factors can be based on facility/site-specific measurements, country specific values or IPCC default values (Table 4.1, chapter 4, Volume 5, 2006 IPCC Guidelines for National Greenhouse Gas Inventories). IPCC default value is 0.01 t CH_4/ t sludge treated on a dry weight basis	tCH_4/t waste treated

6. If the baseline wastewater treatment system is different from the treatment system in the project scenario, the sludge generation rate (amount of sludge generated per unit of COD removed) in the baseline may differ significantly from that of the project scenario. For example, it is known that the amount of sludge generated in aerobic wastewater systems is larger than in anaerobic systems, for the same COD removal efficiency. Therefore, for these cases, the monitored values of the amount of sludge generated during the crediting period will be used to estimate the amount of sludge generated in the baseline, as follows:

$$S_{j,BL,y} = S_{l,PJ,y} \times \frac{SGR_{BL}}{SGR_{PJ}}$$ Equation (5)

Where:

Parameter	Description	Unit
$S_{l,PJ,y}$	Amount of dry matter in the sludge treated by the sludge treatment system *i* in year *y* in the project scenario (t)	t
SGR_{BL}	Sludge generation ratio of the wastewater treatment plant in the baseline scenario (tonne of dry matter in sludge/t COD removed). This ratio will be determined as per paragraphs 35, 36 or 37 of AMS-III.H.	tonne of dry matter in sludge/t COD removed

Parameter	Description	Unit
SGR_{PJ}	Sludge generation ratio of the wastewater treatment plant in the project scenario (tonne of dry matter in sludge/t COD removed). Calculated using the monitored values of COD removal (i.e. $COD_{inflow,i}$ minus $COD_{outflow,i}$) and sludge generation in the project scenario	tonne of dry matter in sludge/t COD removed

7. Methane emissions from degradable organic carbon in treated wastewater discharged in e.g. a river, sea or lake in the baseline situation are determined as follows:

$$BE_{ww,discharge,y} = Q_{ww,y} \times GWP_{CH4} \times B_{o,ww} \times UF_{BL} \times COD_{ww,discharge,BL,y} \times MCF_{ww,BL,discharge} \quad \text{Equation (6)}$$

Where:

Parameter	Description	Unit
$Q_{ww,y}$	Volume of treated wastewater discharged in year y (m^3)	m^3
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	
$COD_{ww,discharge,BL,y}$	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake in the baseline situation in the year y (t/m^3). If the baseline scenario is the discharge of untreated wastewater, the COD of untreated wastewater shall be used	t/m^3
$MCF_{ww,BL,discharge}$	Methane correction factor based on discharge pathway in the baseline situation (e.g. into sea, river or lake) of the wastewater (fraction) (MCF values as per Table 2 of AMS-III.H.)	-

8. To determine $COD_{ww,discharge,BL,y}$: if the baseline treatment system(s) is different from the treatment system(s) in the project scenario, the monitored values of the COD inflow during crediting period will be used to calculate the baseline emissions ex post. The outflow COD of the baseline systems will be estimated using the removal

efficiency of the baseline treatment systems, estimated as per paragraphs 35, 36 or 37 of AMS-III.H.

9. Methane emissions from anaerobic decay of the final sludge produced are determined as follows:

$$BE_{s,final,y} = S_{final,BL,y} \times DOC_s \times UF_{BL} \times MCF_{s,BL,final} \times DOC_F \times F \times 16/12 \times GWP_{CH4} \quad \text{Equation (7)}$$

Where:

Parameter	Description	Unit
$S_{final,BL,y}$	Amount of dry matter in the final sludge generated by the baseline wastewater treatment systems in the year *y* (t). If the baseline wastewater treatment system is different from the project system, it will be estimated using the monitored amount of dry matter in the final sludge generated by the project activity ($S_{final,PJ,y}$) corrected for the sludge generation ratios of the project and baseline systems as per equation (5) above	t
$MCF_{s,BL,final}$	Methane correction factor of the disposal site that receives the final sludge in the baseline situation, estimated as per the procedures described in the methodological tool "Emissions from solid waste disposal sites"	-
UF_{BL}	Model correction factor to account for model uncertainties (0.89)	-

For each PAI in the Grouped Project, the baseline emissions will be identified according to the procedures set out in *AMS-III.H.: Methane recovery in wastewater treatment --- Version 19.0*. The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/ Excluded	Justification
Baseline emissions from electricity or fuel consumption in year *y*	Included/ Excluded	Electricity or fuel may or may not be used in the baseline scenario

Source	Included/ Excluded	Justification
$(BE_{power,y})$		
Baseline emissions of the wastewater treatment systems affected by the project activity in year y $(BE_{ww,treatment,y})$	Included	Main emissions source in the baseline from open lagoons
Baseline emissions of the sludge treatment systems affected by the project activity in year y $(BE_{s,treatment,y})$	Excluded	No sludge treatment activity involved in each PAI
Baseline methane emissions from degradable organic carbon in treated wastewater discharged into sea/river/lake in year y $(BE_{ww,discharge,y})$	Included/ Excluded	The treated effluent from open anaerobic lagoons may or may not be discharged into sea/river/lake
Baseline methane emissions from anaerobic decay of the final sludge produced in year y $(BE_{s,final,y})$	Excluded	No sludge treatment activity involved in each PAI

AMS I D, Version 18.0

According to the approved methodology AMS I D, Version 18.0, the baseline emissions include only CO_2 emissions from electricity generation in power plants that are displaced due to the project activity:

$BE_y = EG_{PJ,y} \times EF_{grid,y}$ Equation (1) of AMS I.D

Where:

Parameter	Description	Unit
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y	MWh
$EF_{grid,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"	tCO_2/MWh

The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/ Excluded	Justification
Baseline emissions from electricity generation in power plants that are displaced due to the project activity (BE_y)	Included/ Excluded	Electricity generation in power plants that are displaced due to the project activity may or may not be applied in the baseline scenario

AMS I F, Version 05.0

According to the approved methodology AMS I F, Version 05.0, the baseline emissions include only CO_2 emissions from electricity generation in power plants that are displaced due to the project activity:

$BE_y = EG_{BLy} \times EF_{CO2,y}$ Equation (2) of AMS I.D

Where:

Parameter	Description	Unit
$EG_{BL,y}$	Quantity of net electricity displaced as a result of the implementation of the CDM project activity in year y	MWh
$EF_{CO2,y}$	Emission factor: • The emission factor of an electric grid shall be calculated as per the procedures provided in AMS-I.D.; • For a mini-grid system other than described in paragraph 19 in AMS ID, the baseline emission factor shall be determined as per the weighted average emissions for the current generation mix following the procedure provided in AMS-I.D • The emission factor for captive electricity generation shall be calculated as per the procedures described in the TOOL05	tCO_2/MWh

The table below summarises the baseline emissions that are considered for each PAI.

Source	Included/ Excluded	Justification
Baseline emissions from amount	Included	Electricity from baseline scenario will

Source	Included/ Excluded	Justification
electricity displaced with the electricity produced by the renewable generating unit (BE_y)		be displaced by the renewable generating unit in project scenario for all the PAIs.

Baseline Emissions for the PAIs

AMS III H, Version 19.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded				Justification
	PAI1 SPMN	PAI2 AAI	PAI3 FDB	PAI4 AWB	
$BE_{power,y}$					No electricity or fossil fuel used in the baseline scenario
$BE_{ww,treatment,y}$	√	√	√	√	Main emissions source in the baseline from open lagoons
$BE_{s,treatment,y}$					No sludge treatment activity involved in all the PAI
$BE_{ww,discharge,y}$		√			Only the effluent from PAI2 is discharged into river, the other PAIs is discharged for land application
$BE_{s,final,y}$					No sludge treatment activity involved in all the PAIs

The summary of baseline emission for all the PAIs is tabulated below:

Year	BE_y	$BE_{ww,treatment,y}$	$BE_{ww,discharge,y}$
01/09/2024–31/08/2024	11,753	11,753	0
01/01/2025–31/12/2025	151,436	151,397	39
01/01/2026–31/12/2026	151,436	151,397	39
01/01/2027–31/12/2027	151,436	151,397	39
01/01/2028–	151,436	151,397	39

Year	BE_y	$BE_{ww,treatment,y}$	$BE_{ww,discharge,y}$
31/12/2028			
01/01/2029–31/12/2029	151,436	151,397	39
01/01/2030–31/12/2030	151,436	151,397	39
01/01/2031–31/08/2031	100,957	100,932	26
Total	**1,021,328**	**1,021,069**	**259**

The baseline emissions calculated for the PAIs is calculated below:

PAI1 - Sarana Prima Multi Niaga (SPMN)

The summary of baseline emission is tabulated below:

Year	BE_y	$BE_{ww,treatment,y}$
01/09/2024– 31/08/2024	11,753	11,753
01/01/2025– 31/12/2025	35,260	35,260
01/01/2026– 31/12/2026	35,260	35,260
01/01/2027– 31/12/2027	35,260	35,260
01/01/2028– 31/12/2028	35,260	35,260
01/01/2029– 31/12/2029	35,260	35,260
01/01/2030– 31/12/2030	35,260	35,260
01/01/2031– 31/08/2031	23,507	23,507
Total	**246,821**	**246,821**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	51,496	154,487	154,487	154,487	154,487	154,487	154,487	102,992
$COD_{inflow,i,y}$	0.0509	0.0509	0.0509	0.0509	0.0509	0.0509	0.0509	0.0509
$h_{COD, BL, i}$	90%	90%	90%	90%	90%	90%	90%	90%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP_{CH4}	28	28	28	28	28	28	28	28

Parameter	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$BE_{ww,treatment,y}$	11,753	35,260	35,260	35,260	35,260	35,260	35,260	23,507

PAI2 - Andalas Agro Industry (AAI)

The summary of baseline emission is tabulated below:

Year	BE_y	$BE_{ww,treatment,y}$	$BE_{ww,discharge,y}$
01/09/2024– 31/08/2024	0	0	0
01/01/2025– 31/12/2025	30,172	30,133	39
01/01/2026– 31/12/2026	30,172	30,133	39
01/01/2027– 31/12/2027	30,172	30,133	39
01/01/2028– 31/12/2028	30,172	30,133	39
01/01/2029– 31/12/2029	30,172	30,133	39
01/01/2030– 31/12/2030	30,172	30,133	39
01/01/2031– 31/08/2031	20,115	20,089	26
Total	**201,147**	**200,888**	**259**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	104,982	104,982	104,982	104,982	104,982	104,982	69,988
$COD_{inflow,i,y}$	0.0587	0.0587	0.0587	0.0587	0.0587	0.0587	0.0587
$h_{COD,BL,i}$	98%	98%	98%	98%	98%	98%	98%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP_{CH4}	28	28	28	28	28	28	28
$BE_{ww,treatment,y}$	**30,133**	**30,133**	**30,133**	**30,133**	**30,133**	**30,133**	**20,089**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$Q_{ww,y}$	104,982	104,982	104,982	104,982	104,982	104,982	69,988
GWP_{CH4}	28	28	28	28	28	28	28
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
COD ww,discharge,BL,y	0.00059	0.00059	0.00059	0.00059	0.00059	0.00059	0.00059
$MCF_{ww,BL,discharge}$	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$BE_{ww,\mathrm{discharge},y}$	**39**	**39**	**39**	**39**	**39**	**39**	**26**

PAI3 - Farinda Bersaudara (FDB)

The summary of baseline emission is tabulated below:

Year	BE_y	$BE_{ww,treatment,y}$
01/09/2024– 31/08/2024	0	0
01/01/2025– 31/12/2025	46,309	46,309
01/01/2026– 31/12/2026	46,309	46,309
01/01/2027– 31/12/2027	46,309	46,309
01/01/2028– 31/12/2028	46,309	46,309
01/01/2029– 31/12/2029	46,309	46,309
01/01/2030– 31/12/2030	46,309	46,309
01/01/2031– 31/08/2031	30,873	30,873
Total	**308,726**	**308,726**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	206,174	206,174	206,174	206,174	206,174	206,174	137,449
$COD_{inflow,i,y}$	0.0501	0.0501	0.0501	0.0501	0.0501	0.0501	0.0501
$h_{COD,BL,i}$	90%	90%	90%	90%	90%	90%	90%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP_{CH4}	28	28	28	28	28	28	28
$BE_{ww,treatment,y}$	**46,309**	**46,309**	**46,309**	**46,309**	**46,309**	**46,309**	**30,873**

PAI4 - Andalas Wahana Berjaya (AWB)

The summary of baseline emission is tabulated below:

Year	BE_y	$BE_{ww,treatment,y}$
01/09/2024– 31/08/2024	0	0
01/01/2025– 31/12/2025	39,695	39,695
01/01/2026– 31/12/2026	39,695	39,695
01/01/2027– 31/12/2027	39,695	39,695
01/01/2028– 31/12/2028	39,695	39,695
01/01/2029– 31/12/2029	39,695	39,695
01/01/2030– 31/12/2030	39,695	39,695
01/01/2031– 31/08/2031	26,463	26,463
Total	**264,635**	**264,635**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	144,647	144,647	144,647	144,647	144,647	144,647	96,432
$COD_{inflow,i,y}$	0.0612	0.0612	0.0612	0.0612	0.0612	0.0612	0.0612
$h_{COD, BL, i}$	90%	90%	90%	90%	90%	90%	90%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
GWP_{CH4}	28	28	28	28	28	28	28
$BE_{ww,treatment,y}$	**39,695**	**39,695**	**39,695**	**39,695**	**39,695**	**39,695**	**26,463**

AMS I D, Version 18.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded				Justification
	PAI1 SPMN	PAI2 AAI	PAI3 FDB	PAI4 AWB	
BE_y	√	√		√	Power generation from the PAIs will be upload to grid except for PAI2. PAI3 will upgrade the biogas to BioCNG

The summary of baseline emission for all the PAIs is tabulated below:

Baseline Emission	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	3,442	18,632	18,632	18,632	18,632	18,632	18,632	12,421	**127,656**

The baseline emissions calculated for the PAIs is calculated below:

PAI1 - Sarana Prima Multi Niaga (SPMN)

The summary of baseline emission is tabulated below:

Baseline Emission	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	3,442	10,327	10,327	10,327	10,327	10,327	10,327	6,885	**72,292**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
EG_{BLy}	2,845	8,535	8,535	8,535	8,535	8,535	8,535	5,690
EF_{CO2}	1.21	1.21	1.21	1.21	1.21	1.21	1.21	1.21
BE_y	3,442	10,327	10,327	10,327	10,327	10,327	10,327	6,885

PAI2 - Andalas Agro Industry (AAI)

The summary of baseline emission is tabulated below:

Baseline Emission	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	**4,434**	**4,434**	**4,434**	**4,434**	**4,434**	**4,434**	**2,956**	**29,558**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
EG_{BLy}	5,278	5,278	5,278	5,278	5,278	5,278	3,519
EF_{CO2}	0.84	0.84	0.84	0.84	0.84	0.84	0.84
BE_y	**4,434**	**4,434**	**4,434**	**4,434**	**4,434**	**4,434**	**2,956**

PAI4 - Andalas Wahana Berjaya (AWB)

The summary of baseline emission is tabulated below:

Baseline Emission	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	**3,871**	**3,871**	**3,871**	**3,871**	**3,871**	**3,871**	**2,581**	**25,805**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
EG_{BLy}	4,608	4,608	4,608	4,608	4,608	4,608	3,072
EF_{CO2}	0.84	0.84	0.84	0.84	0.84	0.84	0.84
BE_y	**3,871**	**3,871**	**3,871**	**3,871**	**3,871**	**3,871**	**2,581**

AMS I F, Version 05.0

The table below summarises the baseline emissions that are considered for the PAIs:

Source	Included/Excluded				Justification
	PAI1 SPMN	PAI2 AAI	PAI3 FDB	PAI4 AWB	
BE_y	√	√	√	√	All the PAIs will displaced the electricity used in baseline with power generated during project activity

The summary of baseline emission for all the PAIs is tabulated below:

Baseline Emission	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	157	2,067	2,067	2,067	2,067	2,067	2,067	1,378	**13,936**

The baseline emissions calculated for the PAIs is calculated below:

PAI1 - Sarana Prima Multi Niaga (SPMN)

The summary of baseline emission is tabulated below:

Baseline Emission	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	157	470	470	470	470	470	470	313	**3,291**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
FC_y	94,773	284,318	284,318	284,318	284,318	284,318	284,318	189,545
$P_{i,y}$	0.865	0.865	0.865	0.865	0.865	0.865	0.865	0.865
FC_y	246	246	246	246	246	246	246	246
$COEF_{CO2}$	1.91	1.91	1.91	1.91	1.91	1.91	1.91	1.91
BE_y	**157**	**470**	**470**	**470**	**470**	**470**	**470**	**313**

PAI2 - Andalas Agro Industry (AAI)

The summary of baseline emission is tabulated below:

Baseline Emission	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	**413**	**413**	**413**	**413**	**413**	**413**	**275**	**2,754**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025 - 31/12/2025	01/01/2026 - 31/12/2026	01/01/2027 - 31/12/2027	01/01/2028 - 31/12/2028	01/01/2029 - 31/12/2029	01/01/2030 - 31/12/2030	01/01/2031 - 31/08/2031
FC_y	249,868	249,868	249,868	249,868	249,868	249,868	166,579
$P_{i,y}$	0.865	0.865	0.865	0.865	0.865	0.865	0.865
FC_y	216	216	216	216	216	216	216
$COEF_{CO2}$	1.91	1.91	1.91	1.91	1.91	1.91	1.91
BE_y	**413**	**413**	**413**	**413**	**413**	**413**	**275**

PAI3 - Farinda Bersaudara (FDB)

The summary of baseline emission is tabulated below:

Baseline Emission	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	**484**	**484**	**484**	**484**	**484**	**484**	**323**	**3,230**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
FC_y	292,967	292,967	292,967	292,967	292,967	292,967	195,311
$P_{i,y}$	0.865	0.865	0.865	0.865	0.865	0.865	0.865
FC_y	253	253	253	253	253	253	253
$COEF_{CO2}$	1.91	1.91	1.91	1.91	1.91	1.91	1.91
BE_y	**484**	**484**	**484**	**484**	**484**	**484**	**323**

PAI4

The summary of baseline emission is tabulated below:

Baseline Emission	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031	Total
BE_y	**699**	**699**	**699**	**699**	**699**	**699**	**466**	**4,661**

The detailed calculation for baseline emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
FC_y	422,854	422,854	422,854	422,854	422,854	422,854	281,903
$P_{i,y}$	0.865	0.865	0.865	0.865	0.865	0.865	0.865
FC_y	366	366	366	366	366	366	366
$COEF_{CO2}$	1.91	1.91	1.91	1.91	1.91	1.91	1.91
BE_y	**699**	**699**	**699**	**699**	**699**	**699**	**466**

4.2 Project Emissions

Project Emissions for the Grouped Project

AMS III H, Version 19.0

According to the approved methodology AMS III H, Version 19.0, the project emissions for instances in the Grouped Project are determined as follows:

10. Project emissions consist of:

    (a) CO_2 emissions from electricity and fuel used by the project facilities ($PE_{power,y}$);

    (b) Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$);

    (c) Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$)

    (d) Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$);

    (e) Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$);

    (f) Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$);

    (g) Methane emissions due to incomplete flaring ($PE_{flaring,y}$);

    (h) Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$);

$$PE_y = \begin{Bmatrix} PE_{power,y} + PE_{ww,treatment,y} + PE_{s,treatment,y} + PE_{ww,discharge,y} + PE_{s,final,y} + \\ PE_{fugitive,y} + PE_{biomass,y} + PE_{flaring,y} \end{Bmatrix} \quad \text{Equation (8)}$$

Where:

Parameter	Description	Unit
PE_y	Project emissions in year y	tCO_2e
$PE_{power,y}$	Emissions from electricity or fuel consumption in the year y (t CO2e). These emissions shall be calculated as per paragraph 26 of AMS III H, for the situation of the project scenario, using energy consumption data of all equipment/devices used in the project activity wastewater and sludge treatment systems and	tCO_2e

Parameter	Description	Unit
	systems for biogas recovery and flaring/gainful use	
$PE_{ww,treatment,y}$	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO_2e). These emissions shall be calculated as per equation (2) in paragraph 27 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project situation ($MCF_{ww,treatment,PJ,k}$ and $\eta_{,PJ,k,y}$) and with the following changed definition of parameters: $MCF_{ww,treatment,PJ,k}$ — Methane correction factor for project wastewater treatment system k (MCF values as per Table 2) $\eta_{,PJ,k,y}$ — Chemical oxygen demand removal efficiency of the project wastewater treatment system k in year y (t/m^3), measured based on inflow COD and outflow COD in system k	tCO_2e
$PE_{s,treatment,y}$	Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO_2e). These emissions shall be calculated as per equations (3) and (4) in paragraphs 30 and 31 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project situation ($S_{l,PJ,y}$, $MCF_{s,treatment,l}$) and with the following changed definition of parameters: $S_{l,PJ,y}$ — Amount of dry matter in the sludge treated by the sludge treatment system l in the project scenario in year y (t) $MCF_{s,treatment,l}$ — Methane correction factor for the project sludge treatment system l (MCF values as per Table 2 of AMS-III.H)	tCO_2e
$PE_{ww,discharge,y}$	Methane emissions from degradable organic carbon in treated wastewater in year y (tCO_2e). These emissions shall be calculated as per equation (6) in	tCO_2e

Parameter	Description	Unit
	paragraph 33 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project conditions ($COD_{ww,discharge,PJ,y}$, $MCF_{ww,PJ,discharge}$) and with the following changed definition of parameters: $COD_{ww,discharge,PJ,y}$ — Chemical oxygen demand of the treated wastewater discharged into the sea, river or lake in the project scenario in year y (t/m^3) $MCF_{ww,PJ,discharge}$ — Methane correction factor based on the discharge pathway of the wastewater in the project scenario (e.g. into sea, river or lake) (MCF values as per Table 2)	
$PE_{s,final,y}$	Methane emissions from anaerobic decay of the final sludge produced in year y ($t\ CO_2e$). These emissions shall be calculated as per equation (7) in paragraph 35 of AMS-III.H, using an uncertainty factor of 1.12 and data applicable to the project conditions ($MCF_{s,PJ,final}$, $S_{final,PJ,y}$). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in aerobic conditions in the project activity, this term shall be neglected, and the sludge treatment and/or use and/or final disposal shall be monitored during the crediting period with the following revised definition of the parameters: $MCF_{s,PJ,final}$ — Methane correction factor of the disposal site that receives the final sludge in the project situation, estimated as per the procedures described in the methodological tool "Emissions from solid waste disposal sites" $S_{final,PJ,y}$ — Amount of dry matter in final sludge generated by the project wastewater treatment	tCO_2e

Parameter	Description	Unit
	systems in the year y (t)	
$PE_{fugitive,y}$	Methane emissions from biogas release in capture systems in year y, calculated as per paragraph 4 (t CO_2e)	tCO_2e
$PE_{biomass,y}$	Methane emissions due to incomplete flaring in year y (t CO_2e). For ex ante estimation, baseline emission calculation for wastewater and/or sludge treatment (i.e. equation (2) and/or equation (3)) can be used but without the consideration of GWP for CH_4. However, the ex post emission reduction shall be calculated as per methodological tool "Project emissions from flaring"	tCO_2e
$PE_{flaring,y}$	Methane emissions from biomass stored under anaerobic conditions. If storage of biomass under anaerobic conditions takes place in the project and does not occur in the baseline, methane emissions due to anaerobic decay of this biomass shall be considered and be determined as per the procedure in the methodological tool "Emissions from solid waste disposal sites" (t CO_2e)	tCO_2e

Project emissions from methane release in capture systems are determined as follows:

Based on the methane emission potential of wastewater and/or sludge:

$$PE_{fugitive,y} = PE_{fugitive,ww,y} + PE_{fugitive,s,y} \qquad \text{Equation (9)}$$

Where:

Parameter	Description	Unit
$PE_{fugitive,ww,y}$	Fugitive emissions through capture inefficiencies in the anaerobic wastewater treatment systems in the year y	tCO_2e
$PE_{fugitive,s,y}$	Fugitive emissions through capture inefficiencies in the anaerobic sludge treatment systems in the year y	tCO_2e

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH4} \qquad \text{Equation (10)}$$

Where:

Parameter	Description	Unit
CFE_{ww}	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)	-
$MEP_{ww,treatment,y}$	Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (t)	-

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_{k} COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k}$$

Equation (11

Where:

Parameter	Description	Unit
$COD_{removed,PJ,k,y}$	The chemical oxygen demand removed[11] by the treatment system k of the project activity equipped with biogas recovery in the year y	t/m^3
$MCF_{ww,treatment,PJ,k}$	Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF values as per Table 2 above)	-
UF_{PJ}	Model correction factor to account for model uncertainties (1.12)	-

$$PE_{fugitive,s,y} = (1 - CFE_s) \times MEP_{s,treatment,y} \times GWP_{CH4}$$

Equation (12)

Where:

Parameter	Description	Unit
CFE_s	Capture efficiency of the biogas recovery equipment in the sludge treatment systems (a default value of 0.9 shall	-

[11] Difference between the inflow COD and the outflow COD.

Parameter	Description	Unit
	be used)	
$MEP_{s,treatment,y}$	Methane emission potential of the sludge treatment systems equipped with a biogas recovery system in year y	t

$$MEP_{s,treatment,y} = \sum_{l} (S_{l,PJ,y} \times MCF_{s,treatment,PJ,l}) \times DOC_s \times UF_{PJ} \times DOC_F \times F \times 16/12 \quad \text{Equation (13)}$$

Where:

Parameter	Description	Unit
$S_{l,PJ,y}$	Amount of sludge treated in the project sludge treatment system l equipped with a biogas recovery system (on a dry basis) in year y	t
$MCF_{s,treatment,PJ,l}$	Methane correction factor for the sludge treatment system equipped with biogas recovery equipment (MCF values as per Table 2 above)	-
UF_{PJ}	Model correction factor to account for model uncertainties (1.12)	-

For each project activity instance in the Grouped Project, the project emissions will be identified according to the procedures set out in *AMS-III.H.: Methane recovery in wastewater treatment --- Version 19.0*. The table below summarises the project emissions that will be considered for each project activity instance in the Grouped Project.

Source	Included/ Excluded	Justification
CO_2 emissions from electricity and fuel used by the project facilities ($PE_{power,y}$)	Included/ Excluded	Electricity or fuel may or may not be used in the project scenario
Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$)	Included/ Excluded	Each project activity instance to determine based on the project design

Source	Included/ Excluded	Justification
Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$)	Excluded	No sludge treatment activity involved in each PAI
Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$)	Included/ Excluded	The treatment effluent from open anaerobic lagoons may or may not be discharged into sea/river/lake
Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$)	Excluded	No sludge treatment activity involved in each PAI
Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$)	Included	Project emissions due to potential inefficiencies of biogas capture system.
Methane emissions due to incomplete flaring ($PE_{flaring,y}$)	Included/ Excluded	The project may or may not emit GHG from incomplete flaring.
Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$)	Excluded	There is no storage of additional biomass under anaerobic conditions in project activity instances.

Project Emissions for the PAIs

AMS III H, Version 19.0

The table below summarises the project emissions that are considered for the PAIs:

Source	Included/Excluded				Justification
	PAI1 SPMN	PAI2 AAI	PAI3 FDB	PAI4 AWB	
$PE_{power,y}$					No electricity or fossil fuel used in the baseline scenario
$PE_{ww,treatment,y}$	√	√	√	√	Main emissions source in the baseline from open lagoons

Source	Included/Excluded				Justification
	PAI1 SPMN	PAI2 AAI	PAI3 FDB	PAI4 AWB	
$PE_{s,treatment,y}$					No sludge treatment activity involved in all the PAI
$PE_{ww,discharge,y}$		√			Only the effluent from PAI2 is discharged into river, the other PAIs is discharged for land application
$PE_{s,final,y}$					No sludge treatment activity involved in all the PAIs
$PE_{fugitive,y}$	√	√	√	√	Project emissions due to potential inefficiencies of biogas capture system
$PE_{flaring,y}$	√	√	√	√	Project emission due to incomplete flaring
$PE_{biomass,y}$					No storage of additional biomass under anaerobic conditions in project activity instances

The summary of project emission for all the PAIs is tabulated below:

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{ww,discharge,y}$	$PE_{fugitive,y}$	$PE_{flaring,y}$
01/09/2024– 31/08/2024	3,912	2,095	0	1,397	420
01/01/2025– 31/12/2025	50,043	26,545	394	17,697	5,407
01/01/2026– 31/12/2026	50,043	26,545	394	17,697	5,407
01/01/2027– 31/12/2027	50,043	26,545	394	17,697	5,407
01/01/2028– 31/12/2028	50,043	26,545	394	17,697	5,407
01/01/2029– 31/12/2029	50,043	26,545	394	17,697	5,407
01/01/2030– 31/12/2030	50,043	26,545	394	17,697	5,407
01/01/2031– 31/08/2031	33,362	17,697	263	11,798	3,605
Total	**337,532**	**179,063**	**2,627**	**119,375**	**36,467**

The project emissions calculated for the PAIs is calculated below:

PAI1 – PT Sarana Prima Multi Niaga (SPMN)

The summary of project emission is tabulated below:

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{fugitive,y}$	$PE_{flaring,y}$
01/09/2024– 31/08/2024	3,912	2,095	1,397	420
01/01/2025– 31/12/2025	11,736	6,286	4,191	1,259

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{\text{fugitive},y}$	$PE_{\text{flaring},y}$
01/01/2026– 31/12/2026	11,736	6,286	4,191	1,259
01/01/2027– 31/12/2027	11,736	6,286	4,191	1,259
01/01/2028– 31/12/2028	11,736	6,286	4,191	1,259
01/01/2029– 31/12/2029	11,736	6,286	4,191	1,259
01/01/2030– 31/12/2030	11,736	6,286	4,191	1,259
01/01/2031– 31/08/2031	7,824	4,191	2,794	840
Total	**82,152**	**44,002**	**29,335**	**8,815**

The detailed calculation for project emission is tabulated below:

Parameter	**01/09/2024-31/12/2024**	**01/01/2025-31/12/2025**	**01/01/2026-31/12/2026**	**01/01/2027-31/12/2027**	**01/01/2028-31/12/2028**	**01/01/2029-31/12/2029**	**01/01/2030-31/12/2030**	**01/01/2031-31/08/2031**
$\Sigma Q_{ww,i,y}$	51,496	154,487	154,487	154,487	154,487	154,487	154,487	102,992
$COD_{removed,PJ,y}$	0.00763	0.00763	0.00763	0.00763	0.00763	0.00763	0.00763	0.00763
h $_{COD,\ PJ,\ i}$	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
$MCF_{ww,treatment,PJ,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28	28
$PE_{ww,treatment,y}$	**2,095**	**6,286**	**6,286**	**6,286**	**6,286**	**6,286**	**6,286**	**4,191**

Parameter	**01/09/2024-31/12/2024**	**01/01/2025-31/12/2025**	**01/01/2026-31/12/2026**	**01/01/2027-31/12/2027**	**01/01/2028-31/12/2028**	**01/01/2029-31/12/2029**	**01/01/2030-31/12/2030**	**01/01/2031-31/08/2031**
CFE_{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP_{CH4}	28	28	28	28	28	28	28	28
$Q_{ww,y}$	51,496	154,487	154,487	154,487	154,487	154,487	154,487	102,992
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
$COD_{removed,PJ,k,y}$	0.0433	0.0433	0.0433	0.0433	0.0433	0.0433	0.0433	0.0433
$MCF_{ww,treatment,PJ,k}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$MEP_{ww,treatment,y}$	1,497	1,497	1,497	1,497	1,497	1,497	1,497	1,497
$PE_{fugitive,y}$	**1,397**	**4,191**	**4,191**	**4,191**	**4,191**	**4,191**	**4,191**	**2,794**

Parameter	01/09/2024-31/12/2024	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	51,496	154,487	154,487	154,487	154,487	154,487	154,487	102,992
$COD_{inflow,i,y}$	0.0509	0.0509	0.0509	0.0509	0.0509	0.0509	0.0509	0.0509
$h_{COD, BL, i}$	90%	90%	90%	90%	90%	90%	90%	90%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
$PE_{flaring,y}$	**420**	**1,259**	**1,259**	**1,259**	**1,259**	**1,259**	**1,259**	**840**

PAI2 – PT Andalas Agro Industry (AAI)

The summary of project emission is tabulated below:

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{ww,discharge,y}$	$PE_{fugitive,y}$	$PE_{flaring,y}$
01/09/2024– 31/08/2024	0	0	0	0	0
01/01/2025– 31/12/2025	9,681	4,927	394	3,284	1,076
01/01/2026– 31/12/2026	9,681	4,927	394	3,284	1,076
01/01/2027– 31/12/2027	9,681	4,927	394	3,284	1,076
01/01/2028– 31/12/2028	9,681	4,927	394	3,284	1,076
01/01/2029– 31/12/2029	9,681	4,927	394	3,284	1,076
01/01/2030– 31/12/2030	9,681	4,927	394	3,284	1,076
01/01/2031– 31/08/2031	6,454	3,284	263	2,190	717
Total	**64,540**	**32,843**	**2,627**	**21,896**	**7,175**

The detailed calculation for project emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	104,982	104,982	104,982	104,982	104,982	104,982	69,988
$COD_{removed,PJ,y}$	0.00880	0.00880	0.00880	0.00880	0.00880	0.00880	0.00880
$h_{COD, PJ, i}$	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
$MCF_{ww,treatment,PJ,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$PE_{ww,treatment,y}$	**4,927**	**4,927**	**4,927**	**4,927**	**4,927**	**4,927**	**3,284**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	104,982	104,982	104,982	104,982	104,982	104,982	69,988
$COD_{ww,discharge,PJ,y}$	0.004787	0.004787	0.004787	0.004787	0.004787	0.004787	0.004787
$MCF_{ww,discharge,PJ,k}$	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28
$PE_{ww,discharge,y}$	**394**	**394**	**394**	**394**	**394**	**394**	**263**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
CFE_{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP_{CH4}	28	28	28	28	28	28	28
$Q_{ww,y}$	104,982	104,982	104,982	104,982	104,982	104,982	69,988
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12
$COD_{removed,PJ,k,y}$	0.0499	0.0499	0.0499	0.0499	0.0499	0.0499	0.0499
$MCF_{ww,treatment,PJ,k}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$MEP_{ww,treatment,y}$	1,173	1,173	1,173	1,173	1,173	1,173	1,173
$PE_{fugitive,y}$	**3,284**	**3,284**	**3,284**	**3,284**	**3,284**	**3,284**	**2,190**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031

$\Sigma Q_{ww,i,y}$	104,982	104,982	104,982	104,982	104,982	104,982	69,988
$COD_{inflow,i,y}$	0.0587	0.0587	0.0587	0.0587	0.0587	0.0587	0.0587
$h_{COD, BL, i}$	98%	98%	98%	98%	98%	98%	98%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
$PE_{flaring,y}$	**1,076**	**1,076**	**1,076**	**1,076**	**1,076**	**1,076**	**717**

PAI3 – PT Farinda Bersaudara (FDB)

The summary of project emission is tabulated below:

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{fugitive,y}$	$PE_{flaring,y}$
01/09/2024– 31/08/2024	0	0	0	0
01/01/2025– 31/12/2025	15,414	8,256	5,504	1,654
01/01/2026– 31/12/2026	15,414	8,256	5,504	1,654
01/01/2027– 31/12/2027	15,414	8,256	5,504	1,654
01/01/2028– 31/12/2028	15,414	8,256	5,504	1,654
01/01/2029– 31/12/2029	15,414	8,256	5,504	1,654
01/01/2030– 31/12/2030	15,414	8,256	5,504	1,654
01/01/2031– 31/08/2031	10,276	5,504	3,669	1,103
Total	**102,757**	**55,039**	**36,692**	**11,026**

The detailed calculation for project emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	206,174	206,174	206,174	206,174	206,174	206,174	137,449
$COD_{removed,PJ,y}$	0.00751	0.00751	0.00751	0.00751	0.00751	0.00751	0.00751
$h_{COD, PJ, i}$	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
$MCF_{ww,treatment,PJ,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
GWP_{CH4}	28	28	28	28	28	28	28
$PE_{ww,treatment,y}$	**8,256**	**8,256**	**8,256**	**8,256**	**8,256**	**8,256**	**5,504**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
CFE_{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP_{CH4}	28	28	28	28	28	28	28
$Q_{ww,y}$	206,174	206,174	206,174	206,174	206,174	206,174	137,449
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12
$COD_{removed,PJ,k,y}$	0.0426	0.0426	0.0426	0.0426	0.0426	0.0426	0.0426
$MCF_{ww,treatment,PJ,k}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$MEP_{ww,treatment,y}$	1,607	1,607	1,607	1,607	1,607	1,607	1,607
$PE_{fugitive,y}$	**5,504**	**5,504**	**5,504**	**5,504**	**5,504**	**5,504**	**3,669**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	206,174	206,174	206,174	206,174	206,174	206,174	137,449
$COD_{inflow,i,y}$	0.0501	0.0501	0.0501	0.0501	0.0501	0.0501	0.0501
$h_{COD, BL, i}$	90%	90%	90%	90%	90%	90%	90%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
$PE_{flaring,y}$	**1,654**	**1,654**	**1,654**	**1,654**	**1,654**	**1,654**	**1,103**

PAI4 – PT Andalas Wahana Berjaya (AWB)

The summary of project emission is tabulated below:

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{fugitive,y}$	$PE_{flaring,y}$
01/09/2024– 31/08/2024	0	0	0	0
01/01/2025– 31/12/2025	13,212	7,077	4,718	1,418

Year	PE_y	$PE_{ww,treatment,y}$	$PE_{\text{fugitive},y}$	$PE_{\text{flaring},y}$
01/01/2026– 31/12/2026	13,212	7,077	4,718	1,418
01/01/2027– 31/12/2027	13,212	7,077	4,718	1,418
01/01/2028– 31/12/2028	13,212	7,077	4,718	1,418
01/01/2029– 31/12/2029	13,212	7,077	4,718	1,418
01/01/2030– 31/12/2030	13,212	7,077	4,718	1,418
01/01/2031– 31/08/2031	8,808	4,718	3,145	945
Total	**88,082**	**47,178**	**31,452**	**9,451**

The detailed calculation for project emission is tabulated below:

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	144,647	144,647	144,647	144,647	144,647	144,647	96,432
$COD_{removed,PJ,y}$	0.00918	0.00918	0.00918	0.00918	0.00918	0.00918	0.00918
$h_{COD, PJ, i}$	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%	85.0%
$MCF_{ww,treatment,PJ,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12
GWP_{CH4}	28	28	28	28	28	28	28
$PE_{ww,treatment,y}$	**7,077**	**7,077**	**7,077**	**7,077**	**7,077**	**7,077**	**4,718**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
CFE_{ww}	0.9	0.9	0.9	0.9	0.9	0.9	0.9
GWP_{CH4}	28	28	28	28	28	28	28
$Q_{ww,y}$	144,647	144,647	144,647	144,647	144,647	144,647	96,432
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{PJ}	1.12	1.12	1.12	1.12	1.12	1.12	1.12
$COD_{removed,PJ,k,y}$	0.0520	0.0520	0.0520	0.0520	0.0520	0.0520	0.0520
$MCF_{ww,treatment,PJ,k}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$MEP_{ww,treatment,y}$	1,007	1,007	1,007	1,007	1,007	1,007	1,007
$PE_{fugitive,y}$	**4,718**	**4,718**	**4,718**	**4,718**	**4,718**	**4,718**	**3,145**

Parameter	01/01/2025-31/12/2025	01/01/2026-31/12/2026	01/01/2027-31/12/2027	01/01/2028-31/12/2028	01/01/2029-31/12/2029	01/01/2030-31/12/2030	01/01/2031-31/08/2031
$\Sigma Q_{ww,i,y}$	144,647	144,647	144,647	144,647	144,647	144,647	96,432
$COD_{inflow,i,y}$	0.0612	0.0612	0.0612	0.0612	0.0612	0.0612	0.0612
$h_{COD, BL, i}$	90%	90%	90%	90%	90%	90%	90%
$MCF_{ww,treatment,BL,i}$	0.8	0.8	0.8	0.8	0.8	0.8	0.8
$B_{o,ww}$	0.25	0.25	0.25	0.25	0.25	0.25	0.25
UF_{BL}	0.89	0.89	0.89	0.89	0.89	0.89	0.89
$PE_{flaring,y}$	**1,418**	**1,418**	**1,418**	**1,418**	**1,418**	**1,418**	**945**

AMS I D, Version 18.0

According to AMS-I.D paragraph 39, the project emission is 0.

AMS I F, Version 05.0

According to AMS-I.F paragraph 26, the project emission is 0.

4.3 Leakage

According to AMS III-H Version 19, paragraph 41, if the technology is using equipment transferred from another activity, leakage effects at the site of the other activity are to be considered and estimated (LE_y). There are no relevant leakage effects to be considered for all the PAIs.

4.4 Net GHG Emission Reductions and Removals

Grouped Project

As per the AMS-III.H. methodology, Version 19.0, emission reductions for the Grouped Project will be calculated as follows.

For all options in paragraph 2 of AMS-III.H, emission reductions shall be estimated ex ante in the PDD using the equations provided in the baseline, project and leakage emissions sections above. Emission reductions shall be estimated ex ante as follows:

$$ER_{y,ex\ ante} = BE_{y,ex\ ante} - \left(PE_{y,ex\ ante} + LE_{y,ex\ ante}\right)$$ Equation (14)

Where:

Parameter	Description	Unit
$ER_{y,ex\ ante}$	Ex-ante emission reduction in year y	tCO_2e
$LE_{y,ex\ ante}$	Ex-ante leakage emissions in year y	tCO_2e
$PE_{y,ex\ ante}$	Ex-ante project emissions in year y calculated as paragraph 39 of AMS-III.H	tCO_2e
$BE_{y,ex\ ante}$	Ex-ante baseline emissions in year y calculated as per paragraph 25 of AMS-III.H	tCO_2e

For cases 2(b), 2(c), 2(d) and 2(f), ex post emission reductions shall be based on the lowest value of the following, as per paragraph 44:

(a) The amount of biogas recovered and fuelled or flared (MD_y) during the crediting period, that is monitored ex post;

(b) Ex post calculated baseline, project and leakage emissions based on actual monitored data for the project activity.

For cases 2(b), 2(c), 2(d) and 2(f): it is possible that the project activity involves wastewater and sludge treatment systems with higher methane conversion factors (*MCF*) or with higher efficiency than the treatment systems used in the baseline situation. Therefore, the emission reductions achieved by the project activity is limited to the ex post calculated baseline emissions minus project emissions using the actual monitored data for the project activity. The emission reductions achieved in any year are the lowest value of the following:

Equation (15)

$$ER_{y,ex\ post} = min\left(\left(BE_{y,ex\ post} - PE_{y,ex\ post} - LE_{y,ex\ post}\right), \left(MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex\ post}\right)\right)$$

Where:

Parameter	Description	Unit
$ER_{y,ex\ post}$	Emission reductions achieved by the project activity based on monitored values for year y	tCO_2e
$BE_{y,ex\ post}$	Baseline emissions calculated as per paragraph 25 of	tCO_2e

Parameter	Description	Unit
	AMS-III.H using ex post monitored values	
$PE_{y,ex\ post}$	Project emissions calculated as per paragraph 39 using ex post monitored values	tCO_2e
MD_y	Methane captured and destroyed/gainfully used by the project activity in the year y	tCO_2e

In the case of flaring/combustion MD_y will be measured using the conditions of the flaring process:

$$MD_y = BG_{burnt,y} \times w_{CH4,y} \times D_{CH4} \times FE \times GWP_{CH4}$$ Equation (16)

Where:

Parameter	Description	Unit
$BG_{burnt,y}$	Biogas flared/combusted in year y	m^3
$w_{CH4,y}$	Methane content of the biogas in the year y	Volume fraction
D_{CH4}	Density of methane at the temperature and pressure of the biogas in the year y	t/m^3
FE	Flare efficiency in year y (fraction). If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied	-

For the cases 2 (a) and (e) the emission reduction achieved by the project activity (ex post) will be the difference between the baseline emissions and the sum of the project emissions and leakage.

$$ER_y = BE_{y,ex\ post} - \left(PE_{y,ex\ post} + LE_{y,ex\ post}\right)$$ Equation (17)

The historical records of electricity and fuel consumption, the COD content of untreated and treated wastewater, and the quantity of sludge produced by the replaced units will be used for the baseline calculation.

In case (a), if the volumetric flow and the characteristic properties (e.g. COD) of the inflow and outflow of the wastewater are equivalent in the project and the baseline scenarios (i.e. the project and baseline systems have the same efficiency for COD removal for wastewater treatment), then the higher energy consumption and sludge generation in the baseline scenario are the only significant differences contributing to emissions reductions in the project case. In this case, the emission reductions can be calculated as the difference between the historical energy consumption of the replaced unit and the recorded energy consumption of the new system, plus the difference in emissions from sludge treatment and/or disposal. Project emissions from fugitive emissions and incomplete flaring ($PE_{fugitive,y}$, $PE_{flaring,y}$) shall also be considered in the calculation of the emission reductions, however the emissions from the wastewater outflow and sludge ($PE_{ww,discharge,y}$, $PE_{s,final,y}$) may be disregarded, if they are equivalent in the baseline and project scenarios.

Grouped Project

The estimated emissions reduction for all the PAIs are calculated as per Equation 14 of AMS-III.H, Version 19.0, which is tabulated as below:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01/09/2024 – 31/12/2024	15,353	3,912	0	11,441
01/01/2025 – 31/12/2025	172,135	50,043	0	122,092
01/01/2026 – 31/12/2026	172,135	50,043	0	122,092
01/01/2027 – 31/12/2027	172,135	50,043	0	122,092
01/01/2028 – 31/12/2028	172,135	50,043	0	122,092
01/01/2029 – 31/12/2029	172,135	50,043	0	122,092
01/01/2030 – 31/12/2020	172,135	50,043	0	122,092

01/01/2031 – 31/08/2031	114,757	33,362	0	81,395
Total	**1,162,920**	**337,532**	**0**	**825,388**

PAI1 – PT Sarana Prima Multi Niaga (SPMN)

The summary of emission reduction is tabulated below:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01/09/2024 – 31/12/2024	15,353	3,912	0	11,441
01/01/2025 – 31/12/2025	46,058	11,736	0	34,322
01/01/2026 – 31/12/2026	46,058	11,736	0	34,322
01/01/2027 – 31/12/2027	46,058	11,736	0	34,322
01/01/2028 – 31/12/2028	46,058	11,736	0	34,322
01/01/2029 – 31/12/2029	46,058	11,736	0	34,322
01/01/2030 – 31/12/2020	46,058	11,736	0	34,322
01/01/2031 – 31/08/2031	30,705	7,824	0	22,881
Total	**322,404**	**82,152**	**0**	**240,251**

PAI2 – PT Andalas Agro Industry (AAI)

The summary of emission reduction is tabulated below:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01/09/2024 – 31/12/2024	0	0	0	0
01/01/2025 – 31/12/2025	35,019	9,681	0	25,338
01/01/2026 – 31/12/2026	35,019	9,681	0	25,338
01/01/2027 – 31/12/2027	35,019	9,681	0	25,338
01/01/2028 – 31/12/2028	35,019	9,681	0	25,338

01/01/2029 – 31/12/2029	35,019	9,681	0	25,338
01/01/2030 – 31/12/2020	35,019	9,681	0	25,338
01/01/2031 – 31/08/2031	23,346	6,454	0	16,892
Total	**233,460**	**64,540**	**0**	**168,919**

PAI3 – PT Farinda Bersaudara (FDB)

The summary of emission reduction is tabulated below:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01/09/2024 – 31/12/2024	0	0	0	0
01/01/2025 – 31/12/2025	46,793	15,414	0	31,380
01/01/2026 – 31/12/2026	46,793	15,414	0	31,380
01/01/2027 – 31/12/2027	46,793	15,414	0	31,380
01/01/2028 – 31/12/2028	46,793	15,414	0	31,380
01/01/2029 – 31/12/2029	46,793	15,414	0	31,380
01/01/2030 – 31/12/2020	46,793	15,414	0	31,380
01/01/2031 – 31/08/2031	31,196	10,276	0	20,920
Total	**311,955**	**102,757**	**0**	**209,198**

PAI4 – PT Andalas Wahana Berjaya (AWB)

The summary of emission reduction is tabulated below:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
01/09/2024 – 31/12/2024	0	0	0	0
01/01/2025 – 31/12/2025	44,265	13,212	0	31,053
01/01/2026 – 31/12/2026	44,265	13,212	0	31,053

01/01/2027 – 31/12/2027	44,265	13,212	0	31,053
01/01/2028 – 31/12/2028	44,265	13,212	0	31,053
01/01/2029 – 31/12/2029	44,265	13,212	0	31,053
01/01/2030 – 31/12/2020	44,265	13,212	0	31,053
01/01/2031 – 31/08/2031	29,510	8,808	0	20,702
Total	**295,102**	**88,082**	**0**	**207,020**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$MCF_{ww,treatment,BL,i}$
Data unit	-
Description	*Methane correction factor for baseline wastewater treatment systems (Anaerobic deep lagoon - depth more than 2 metres)*
Source of data	*MCF value as per table 2. IPCC default values for Methane Correction Factor (MCF)*
Value applied	*0.8*
Justification of choice of data or description of measurement methods and procedures applied	*Anaerobic deep lagoon - depth more than 2 metres*
Purpose of Data	*Calculation of baseline emission*
Comments	*Applicable to PAI1, PAI2, PAI3 and PAI4*

Data / Parameter	$MCF_{ww,BL,discharge}$
Data unit	-
Description	*Methane correction factor for baseline wastewater treatment systems (discharge of wastewater to river)*
Source of data	*MCF value as per table 2, AMS-III. H (Version 19)*
Value applied	*0.1*
Justification of choice of data or description of measurement methods and	*Discharge of wastewater to river*

procedures applied	
Purpose of Data	*Calculation of baseline emission*
Comments	*Applicable to PAI2*

Data / Parameter	$B_{o,ww}$
Data unit	-
Description	*Methane producing capacity of the wastewater*
Source of data	*Default value as per methodology AMS-III. H (Version 19)*
Value applied	*0.25*
Justification of choice of data or description of measurement methods and procedures applied	*Default value as per methodology AMS-III. H (Version 19)*
Purpose of Data	*Calculation of baseline and project emissions*
Comments	-

Data / Parameter	UF_{BL}
Data unit	-
Description	*Model correction factor to account for model uncertainties*
Source of data	*Default value as per methodology AMS-III. H (Version 19)*
Value applied	*0.89*
Justification of choice of data or description of measurement methods and procedures applied	*Correction factor for uncertainties*
Purpose of Data	*Calculation of baseline emissions*

Comments	-

Data / Parameter	GWP_{CH4}
Data unit	tCO_2/tCH_4
Description	*Global Warming Potential for methane*
Source of data	*IPCC*
Value applied	*28*
Justification of choice of data or description of measurement methods and procedures applied	*Default value as per IPCC*
Purpose of Data	*Calculation of baseline and project emissions*
Comments	-

Data / Parameter	$MCF_{ww,treatment,PJ,i}$
Data unit	-
Description	*Methane correction factor for project wastewater treatment systems j*
Source of data	*MCF value as per table 2, AMS-III. H (Version 19)*
Value applied	*0.8*
Justification of choice of data or description of measurement methods and procedures applied	*Project wastewater system without biogas recovery is discharged to lagoons of depth more than 2 metre*
Purpose of Data	*Calculation of project emission*
Comments	*Applicable to PAI1, PAI2, PAI3 and PAI4*

Data / Parameter	$MCF_{ww,discharge,PJ,k}$
Data unit	-
Description	*Methane correction factor based on the discharge pathway of the wastewater in the project scenario*
Source of data	*MCF value as per table 2, AMS-III. H (Version 19)*
Value applied	*0.1*
Justification of choice of data or description of measurement methods and procedures applied	Discharge of wastewater to river for PAI3. *Land application for PAI1, PAI2 and PAI4.*
Purpose of Data	*Calculation of project emissions.*
Comments	-

Data / Parameter	UF_{PJ}
Data unit	-
Description	*Model correction factor to account for model uncertainties*
Source of data	*Default value as per methodology AMS-III. H (Version 19)*
Value applied	*0.89*
Justification of choice of data or description of measurement methods and procedures applied	*Correction factor for uncertainties*
Purpose of Data	*Calculation of project emissions.*
Comments	-

Data / Parameter	CFE_{ww}
Data unit	-
Description	*Capture efficiency of the biogas recovery equipment in the wastewater treatment systems*
Source of data	*Default value as per methodology AMS-III. H (Version 19)*
Value applied	*0.9*
Justification of choice of data or description of measurement methods and procedures applied	*Default value as per methodology AMS-III. H (Version 19)*
Purpose of Data	*Calculation of project emissions*
Comments	-

Data / Parameter	$P_{i,y}$
Data unit	*kg/litre*
Description	*Biodiesel density*
Source of data	*https://www.cpopc.org/wp-content/uploads/2022/04/2-1-LEMIGAS-Palm-Biodiesel-Conference-2022-compressed.pdf*
Value applied	*0.865*
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	*For baseline and project emissions*
Comments	*Applicable for all PAIs*

Data / Parameter	$NVC_{diesel,y}$

Data unit	*GJ/ton*
Description	*Weighted average net calorific value of biodiesel*
Source of data	2006 IPCC
Value applied	*27*
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	*For calculation of baseline emission*
Comments	*Applicable for all PAIs*

Data / Parameter	$EF_{CO2,diesel,y}$
Data unit	*ton/GJ*
Description	*Weighted average* CO_2 *emission factor of biodiesel*
Source of data	2006 IPCC
Value applied	*0.0708*
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	*For calculation of project emission*
Comments	*Applicable for all PAIs*

Data / Parameter	$EF_{CO2,kaltim\ grid,i}$
Data unit	$kgCO_2/MWh$
Description	*Grid emission factor for Kalimantan Timur grid*

Source of data	*Published by Ministry of Energy and Mineral Resources, Indonesia* https://jdih.esdm.go.id/storage/document/Kepmen%20ESDM%20No.%20163.K_HK.02-MEM.S-2021.pdf
Value applied	*1.14*
Justification of choice of data or description of measurement methods and procedures applied	*Ex-ante value, determined by the Ministry*
Purpose of Data	*For PAI3*
Comments	*Published by Ministry of Energy and Mineral Resources, Indonesia*

Data / Parameter	$EF_{CO2,kalteng\ grid,i}$
Data unit	$kgCO_2/MWh$
Description	*Grid emission factor for Kalimantan Tengah grid*
Source of data	*Published by Ministry of Energy and Mineral Resources, Indonesia* https://jdih.esdm.go.id/storage/document/Kepmen%20ESDM%20No.%20163.K_HK.02-MEM.S-2021.pdf
Value applied	*1.28*
Justification of choice of data or description of measurement methods and procedures applied	*Ex-ante value, determined by the Ministry*
Purpose of Data	*For PAI1*
Comments	*Published by Ministry of Energy and Mineral Resources, Indonesia*

Data / Parameter	$EF_{CO2,sumbar\ grid,i}$
Data unit	$kgCO_2/MWh$
Description	*Grid emission factor for Sumatra Barat grid*
Source of data	*Published by Ministry of Energy and Mineral Resources, Indonesia* https://jdih.esdm.go.id/storage/document/Kepmen%20ESDM%20No.%20163.K_HK.02-MEM.S-2021.pdf
Value applied	*0.93*
Justification of choice of data or description of measurement methods and procedures applied	*Ex-ante value, determined by the Ministry*
Purpose of Data	*For PAI2 and PAI4*
Comments	*Published by Ministry of Energy and Mineral Resources, Indonesia*

Data / Parameter	$\eta_{COD,\ BL,}$
Data unit	*%*
Description	*COD removal efficiency of the baseline system i*
Source of data	*Based on historical data*
Value applied	*PAI1: 90%* *PAI2: 98%* *PAI3: 90%* *PAI4: 90%*
Justification of choice of data or description of measurement methods and procedures applied	-

Purpose of Data	*For calculation of baseline emission*
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$Q_{ww,y}$
Data unit	m^3
Description	*Volume of wastewater treated in the project wastewater system in year y*
Source of data	*On-site measurement*
Description of measurement methods and procedures to be applied	*Measured continuously using flow meter installed at entry of the system*
Frequency of monitoring/recording	*Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)*
Value applied	*PAI1: 154,487 m^3* *PAI2: 104,982 m^3* *PAI3: 206,174 m^3* *PAI4: 144,647 m^3*
Monitoring equipment	*Flow meter*
QA/QC procedures to be applied	*Flow meters will undergo maintenance/calibration subject to appropriate industry standards. If local/national standards or the manufacturer's specifications are not available, international standards may be used.*
Purpose of data	*Calculation of baseline and project emissions*
Calculation method	-
Comments	-

Data / Parameter	*$COD_{ww,untreated,PJ,y}$ or $COD_{inflow, PJ,y}$*
Data unit	*t/m^3*
Description	*Chemical oxygen demand of the wastewater entering the biogas treatment system in year y (tonnes/m^3)*
Source of data	*On-site measurement*
Description of measurement methods and procedures to be applied	*Measured the COD according to national or international standards. COD is measured through international sampling*
Frequency of monitoring/recording	*Periodically, at least monthly*
Value applied	*PAI1: 0.0509 t/m^3* *PAI2: 0.0587 t/m^3* *PAI3:0.0501 t/m^3* *PAI4:0.0612 t/m^3*
Monitoring equipment	*Laboratory analysis*
QA/QC procedures to be applied	*Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level*
Purpose of data	*Calculation of baseline and project emissions*
Calculation method	-
Comments	-

Data / Parameter	*$COD_{ww,treated,PJ,y}$ or $COD_{outflow, PJ,y}$*
Data unit	*t/m^3*
Description	*Chemical oxygen demand of the wastewater leaving the biogas treatment system in the year y (tonnes/m3)*
Source of data	*On-site measurement*
Description of measurement	*Measured the COD according to national or international*

methods and procedures to be applied	*standards. COD is measured through international sampling*
Frequency of monitoring/recording	*Periodically, at least monthly*
Value applied	*Based on 85% COD removal efficiency of bio-digester:* *PAI1: 0.0433 t/m³* *PAI2: 0.0499 t/m³* *PAI3:0.0426 t/m³* *PAI4:0.0520 t/m³*
Monitoring equipment	*Laboratory analysis*
QA/QC procedures to be applied	*Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level*
Purpose of data	*Calculation of baseline and project emissions*
Calculation method	-
Comments	-

Data / Parameter	$COD_{ww,discharge,PJ,y}$
Data unit	*t/m^3*
Description	*Chemical oxygen demand of the wastewater outflow to the river in year y (tonnes/m^3)*
Source of data	*On-site measurement*
Description of measurement methods and procedures to be applied	*Measured the COD according to national or international standards. COD is measured through international sampling*
Frequency of monitoring/recording	*Periodically, at least monthly*
Value applied	*Based on COD removal efficiency of the project wastewater treatment system which is not equipped with biogas recovery:*

	PAI1: 100% *PAI2: 46%* *PAI3: 100%* *PAI4: 46%*
Monitoring equipment	*Laboratory analysis*
QA/QC procedures to be applied	*Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level*
Purpose of data	*Calculation of baseline and project emissions*
Calculation method	-
Comments	-

Data / Parameter	$COD_{removed,PJ,y}$
Data unit	*tonne/m³*
Description	*Remaining of the COD post biogas digesters in year y*
Source of data	*Calculated from :* $(COD_{ww,untreated,PJ,y})$-($COD_{ww,treated,PJ,y})$
Description of measurement methods and procedures to be applied	*Measured the COD according to national or international standards. COD is measured through international sampling*
Frequency of monitoring/recording	*Periodically, at least monthly*
Value applied	*PAI1: 0.0076 tonne/m³* *PAI2: 0.0088 tonne/m³* *PAI3: 0.0075 tonne/m³* *PAI4: 0.0092 tonne/m³*
Monitoring equipment	-
QA/QC procedures to be applied	*Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and*

	measurements shall ensure a 90/10 confidence level
Purpose of data	*Calculation of baseline and project emissions*
Calculation method	-
Comments	-

Data / Parameter	$\eta_{COD, PJ,y}$
Data unit	*%*
Description	*COD removal efficiency of biogas digester in project case*
Source of data	*Calculated*
Description of measurement methods and procedures to be applied	$(COD_{outflow,y} - COD_{inflow,y}) / COD_{outflow,y}$
Frequency of monitoring/recording	*Monthly*
Value applied	*85% (as per default value of methodology)*
Monitoring equipment	*Calculated*
QA/QC procedures to be applied	-
Purpose of data	*Calculation of baseline and project emissions*
Calculation method	-
Comments	-

Data / Parameter	$S_{final,PJ,y}$
Data unit	*tonne*

Description	*Method of disposal of the sludge*
Source of data	*Record of site sludge disposal*
Description of measurement methods and procedures to be applied	*Use of sludge for land disposal is monitored*
Frequency of monitoring/recording	*At each incidence of sludge disposal*
Value applied	*For all the PAIs, the estimated value for both ex-ante and ex-post calculation is 0 tonne*
Monitoring equipment	*Flow meter*
QA/QC procedures to be applied	*Records are approved and signed by plant manager*
Purpose of data	*Monitoring of sludge disposal*
Calculation method	-
Comments	*Sludge is land applied.*

Data / Parameter	$FV_{flared,h}$
Data unit	*Nm 3 / hour*
Description	*Volumetric flow of biogas recovered and flared at normal conditions in the hour h*
Source of data	*Measured continuously*
Description of measurement methods and procedures to be applied	*The volume of biogas will be measured with a normalised flow meter or calculated from the volumetric flow, pressure, temperature*
Frequency of monitoring/recording	*Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)*
Value applied	*Values to be monitored ex-post*

Monitoring equipment	*Flow meter(s)*
QA/QC procedures to be applied	*The biogas flow is monitored continuously, recorded and stored electronically in a data log file (DLF) (monthly or at shorter interval). Maintenance and calibration as per manufacturer's specifications or at least once every 3 years*
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	-

Data / Parameter	$FV_{fuelled,h}$
Data unit	*Nm³ / hour*
Description	*Volumetric flow of biogas recovered and fueled at normal conditions in the hour h*
Source of data	*Measured continuously*
Description of measurement methods and procedures to be applied	*The volume of biogas will be measured with a normalised flow meter or calculated from the volumetric flow of biogas, pressure, temperature*
Frequency of monitoring/recording	*Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)*
Value applied	*Values to be monitored ex-post*
Monitoring equipment	*Flow meter(s)*
QA/QC procedures to be applied	*The biogas flow is monitored continuously, recorded and stored electronically in a data log file (DLF) (monthly or at shorter interval).* *Maintenance and calibration as per manufacturer's specifications or at least once every 3 years*

Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	-

Data / Parameter	$fv_{CH4,h}$
Data unit	*%*
Description	*Methane content in biogas in year y*
Source of data	*Continuous measurement*
Description of measurement methods and procedures to be applied	*Using calibrated gas analyzer either continuous or periodical measurement at confidence/precision level of 90/10.*
Frequency of monitoring/recording	*Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)*
Value applied	*Values to be monitored ex-post*
Monitoring equipment	*Gas analyzer*
QA/QC procedures to be applied	*In case a continuous gas analyser is used, data will be recorded and stored electronically in a data log file (DLF) every hour or at shorter interval.* *Maintenance and calibration as per manufacturer's specifications or at least once every 3 years.*
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	-

Data / Parameter	*T_{biogas}*
Data unit	*°C*
Description	*Temperature of biogas*
Source of data	*On-line monitoring*
Description of measurement methods and procedures to be applied	*Pressure and temperature are required to determine the density of the methane. If the biogas flow meter employed measures flow, pressure and temperature and displays/outputs normalised flow of biogas, there is no need to separate monitoring of pressure and temperature of the biogas.*
Frequency of monitoring/recording	*Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)*
Value applied	*Value to be monitored ex-post*
Monitoring equipment	*Temperature probe*
QA/QC procedures to be applied	*Shall be measured and recorded at the same time when methane content in biogas is measured.* *Maintenance and calibration as per manufacturer's specifications or at least once every 3 years*
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	*Pressure and temperature are required to determine the density of the methane. If the biogas flow meter employed measures flow, pressure and temperature and displays/outputs normalised flow of biogas, there is no need to separate monitoring of pressure and temperature of the biogas.*

Data / Parameter	*P_{biogas}*
Data unit	*Bar (g)*

Description	*Pressure of biogas*
Source of data	*On-line measurement*
Description of measurement methods and procedures to be applied	*Pressure measurement device.*
Frequency of monitoring/recording	*Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)*
Value applied	*Value to be monitored ex-post*
Monitoring equipment	*Pressure measurement device.*
QA/QC procedures to be applied	*Shall be measured and recorded at the same time when methane content in biogas is measured.* *Maintenance and calibration as per manufacturer's specifications or at least once every 3 years*
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	*Pressure and temperature are required to determine the density of the methane. If the biogas flow meter employed measures flow, pressure and temperature and displays/outputs normalised flow of biogas, there is no need to separate monitoring of pressure and temperature of the biogas.*

Data / Parameter	$\eta_{flare,h}$
Data unit	*%*
Description	*Flare efficiency in hour h*
Source of data	*Calculated*
Description of measurement methods and procedures to	*This efficiency will be based on hourly flare efficiencies calculated as per the provisions in the "Tool to determine*

be applied	*project emissions from flaring gases containing methane". The hourly flare efficiency will be calculated based on the duration of time during which the flame is detected, by the thermocouple (or flame detector) installed in the open flare. Flare efficiency in hour h is based on default values:* *• 0% if the flame detected for not more than 20 minutes during the hour h and;* *• 50% if the flame detected for more than 20 minutes during the hour h*
Frequency of monitoring/recording	*Hourly*
Value applied	*Value to be monitored ex-post*
Monitoring equipment	*A flame detector (thermocouple or equivalent) will be used.*
QA/QC procedures to be applied	*Data are logged hourly in a data log file (DLF).*
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	*The only monitored parameter related to the flare efficiency is flame detection.*

Data / Parameter	*Flame detector*
Data unit	*On/off*
Description	*Flame detection*
Source of data	*On-line*
Description of measurement methods and procedures to be applied	*Detection of flame*
Frequency of	*Hourly*

monitoring/recording	
Value applied	*On-off*
Monitoring equipment	*A flame detector (thermocouple or equivalent) will be used.*
QA/QC procedures to be applied	*Data are logged hourly in a data log file (DLF).*
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	*For determination of flare efficiency ($\eta_{flare,h}$)*

Data / Parameter	$EC_{PJ,y}$
Data unit	*MWh/year*
Description	*Electricity consumption in the project scenario in the year y*
Source of data	*On-site measurement*
Description of measurement methods and procedures to be applied	*Measurement by electricity meters on the electricity entering the Biogas Plant (from the Mill).*
Frequency of monitoring/recording	*Continuously, ongoing.*
Value applied	*Value to be monitored ex-post*
Monitoring equipment	*Electricity meter*
QA/QC procedures to be applied	*maintenance and calibration of the meters according to the manufacturer's recommendations, industry standard or on the national grid operator's requirement and standard practice.* *The meter data can be cross-checked against the electricity sales invoices presented to the Biogas plant form the electricity providers.*

Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	*For the Grouped Project, measurement with electricity meter(s).* *For PAI1, as the electricity consumption will be supplied from the mill, the electricity consumption will be recorded from the electricity meter owned by the mill.*

Data / Parameter	$FC_{diesel,y}$
Data unit	*litre*
Description	*Diesel consumption in project activity in year, y*
Source of data	*On-site logbook*
Description of measurement methods and procedures to be applied	*On-site measurement*
Frequency of monitoring/recording	*Continuous monitoring and monthly compilation*
Value applied	*Value to be monitored ex-post*
Monitoring equipment	*Metering*
QA/QC procedures to be applied	-
Purpose of data	*Calculation of project emissions*
Calculation method	-
Comments	-

5.3 Monitoring Plan

The purpose of monitoring plan is to ensure that the required data are accurately monitored and recorded to enable the calculation of the emission reductions achieved by the PAI's.

All the PAI's will have similar structure of monitoring and management team.

Monitoring and management systems:

Position	***Outline of responsibilities***	***Reporting***
Mill Manager/ Assistant Mill Manager	• *Reviews the monthly reports and investigates any irregularities* • *Ensure on-going compliance with VCS monitoring plan* • *Supervise meter calibration requirements*	*Management Team*
Supervisor/Officer	• *Oversees collection, recording and storage of data* • *Responsible for ensuring data recording and meters are functioning correctly*	*Mill Manager/ Assistant Mill Manager*
Plant Operator (Shift based)	• *The person appointed for each shift involves in operating and maintaining the plant* • *Responsible for data recording and checking meter functions* • *Responsible for collection of wastewater samples for the purpose of the COD measurement*	*Supervisor / Officer*
Engine driver/ Charge-man	• *The person appointed must be an experienced person to operate and maintain the biogas engines*	*Mill Manager/ Assistant Mill Manager*

Training:

Special training will be provided to the biogas plant operators by the technology provider on the operational and maintenance procedures after the testing and commissioning period. A CDM monitoring Manual will be drafted and brief training will be provided by an appointed VCS consultant on the monitoring parameters.

Emergency procedures for the monitoring system:

1. *Data collection*

   If the person-in-charge for daily raw data logging is absent, there will be another back-up person who takes over the job.

2. *Data storage, transfer and backup*

   The monitored parameters are collected and stored in the following ways:

   (a) Hard copy

   - *Log sheets/ log books will be kept in proper manner*

   (b) Soft copy

   - *Raw data from logsheets/ log books will be stored as soft copy; together with monthly CER calculation, the entire CDM data will be back-up to an external hard drive or another computer every quarterly.*

APPENDIX

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.